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(54) METHOD OF FABRICATING SOI DEVICE WITH CARBON IN BODY REGIONS

(71) Applicant: **Newport Fab, LLC dba Tower Semiconductor Newport Beach**, Newport Beach, CA (US)

(72) Inventor: **Kurt Moen**, Newport Beach, CA (US)

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H10D 86/01	(2025.01)

(52) **U.S. Cl.**

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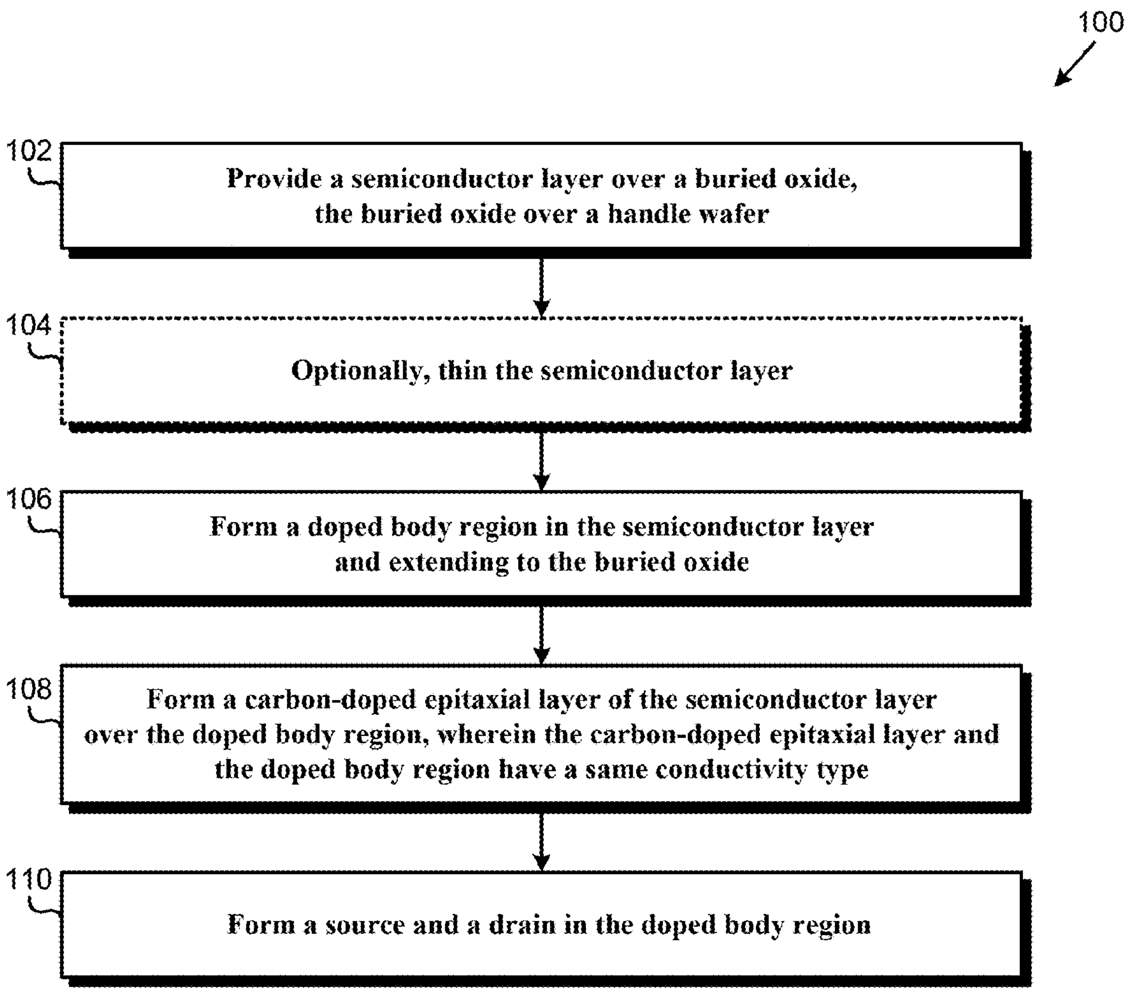
(57) **ABSTRACT**

A semiconductor-on-insulator (SOI) structure includes a semiconductor layer over a buried oxide over a handle wafer. A carbon-doped epitaxial layer is in the semiconductor layer. A doped body region is in the semiconductor layer under the carbon-doped epitaxial layer and extending to the buried oxide. The carbon-doped epitaxial layer and the doped body region have a same conductivity type. Alternatively, a doped body region in the semiconductor layer and extending to the buried oxide includes carbon dopants and body dopants, wherein a peak carbon dopant concentration is situated at a first depth, and a peak body dopant concentration is situated at a second depth below the first depth. Alternatively, an SOI transistor in the semiconductor layer includes a halo region having a different conductivity type from a source and a drain. The halo region includes carbon dopants and body dopants. The source and/or the drain adjoin the halo region.



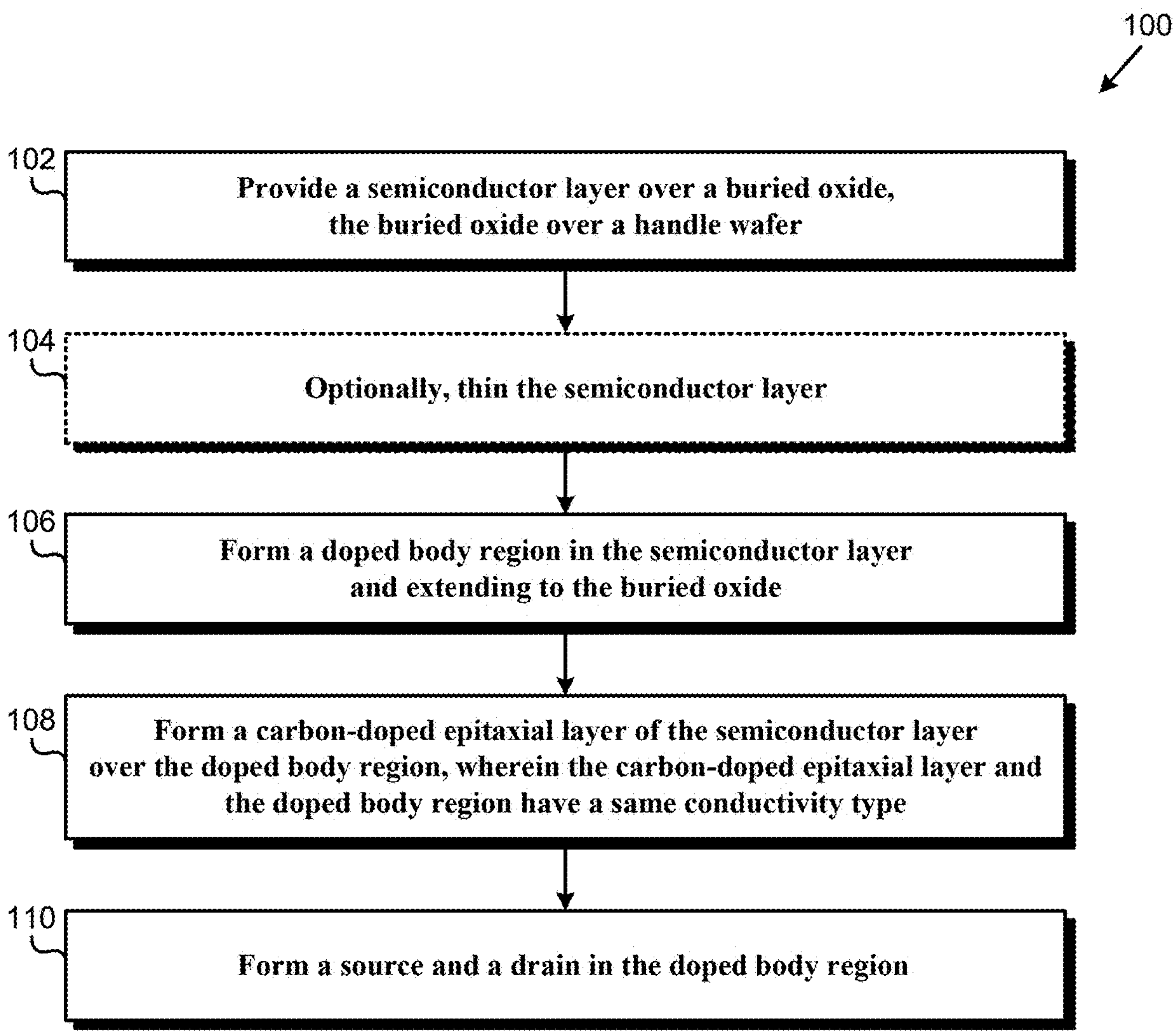

100
102
Provide a semiconductor layer over a buried oxide,
the buried oxide over a handle wafer
104
Optionally, thin the semiconductor layer
106
Form a doped body region in the semiconductor layer
and extending to the buried oxide
108
Form a carbon-doped epitaxial layer of the semiconductor layer
over the doped body region, wherein the carbon-doped epitaxial layer and
the doped body region have a same conductivity type
110
Form a source and a drain in the doped body region

FIG. 1

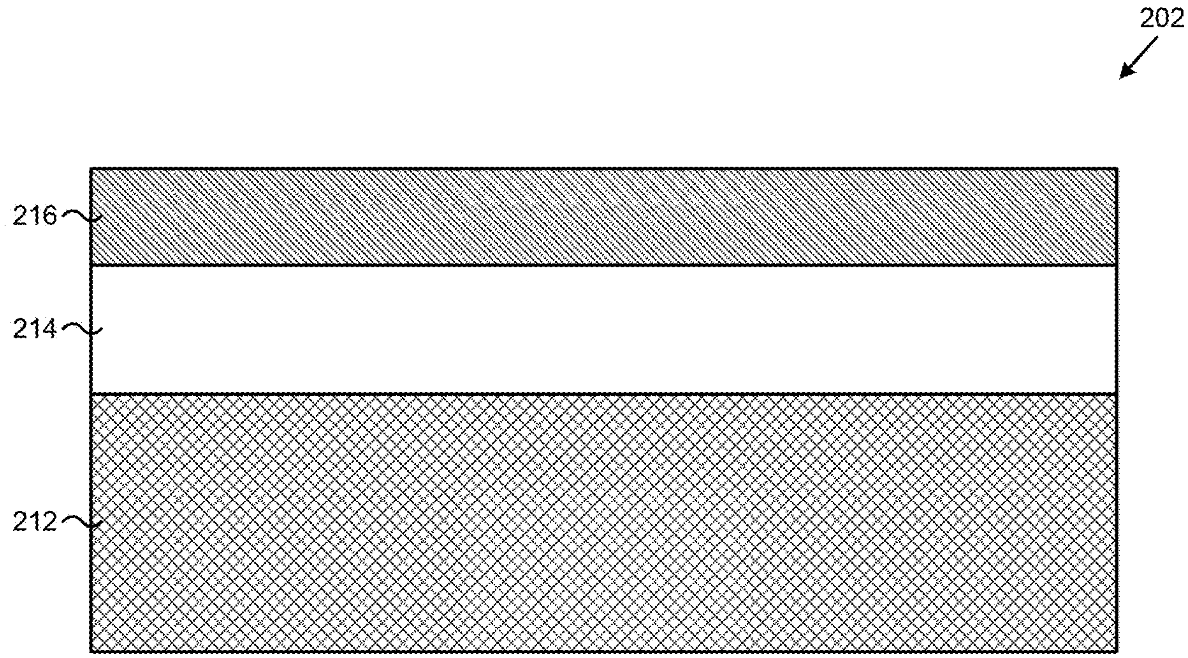


FIG. 2

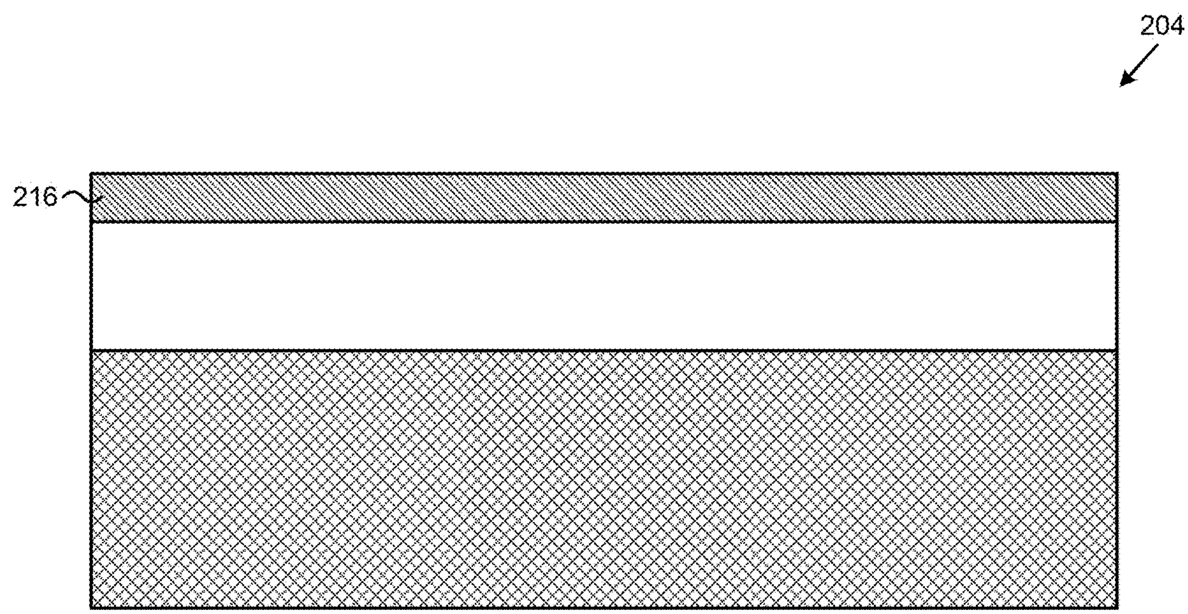


FIG. 3

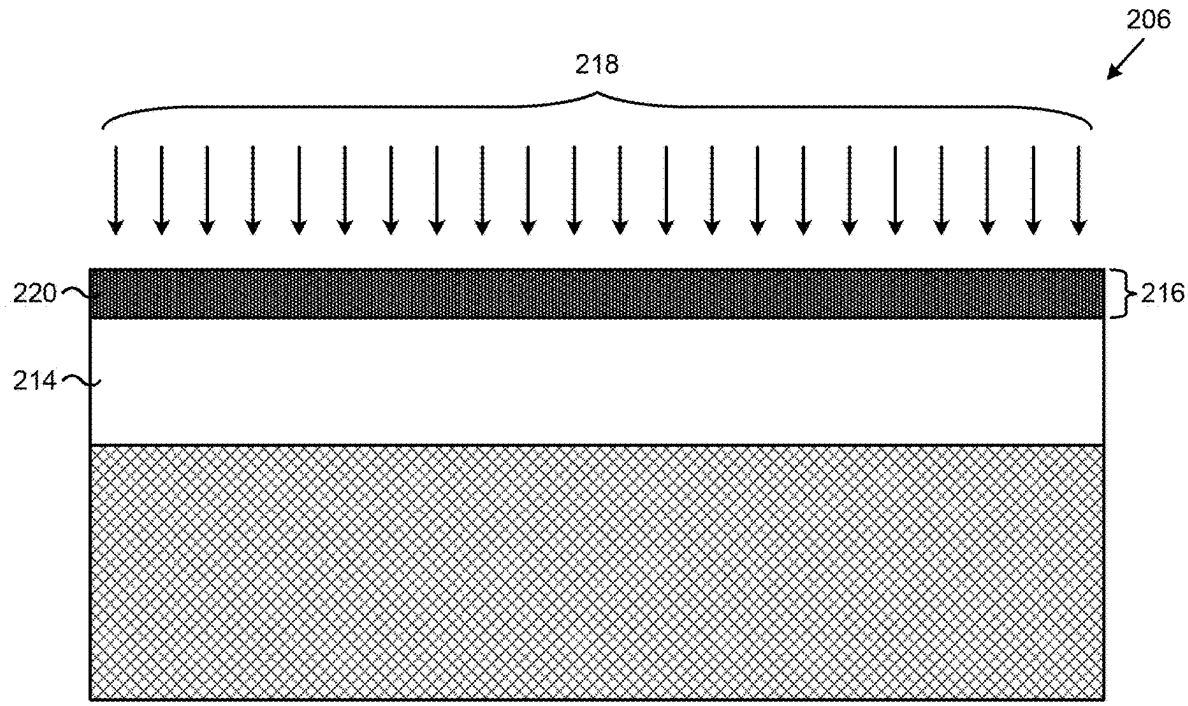


FIG. 4

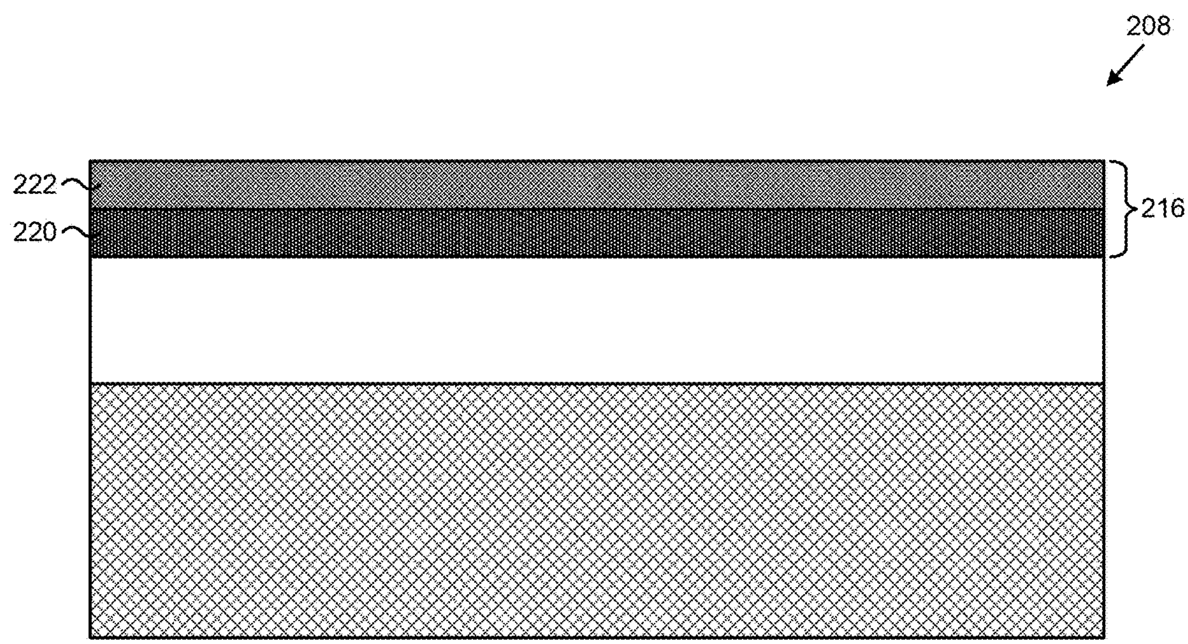


FIG. 5

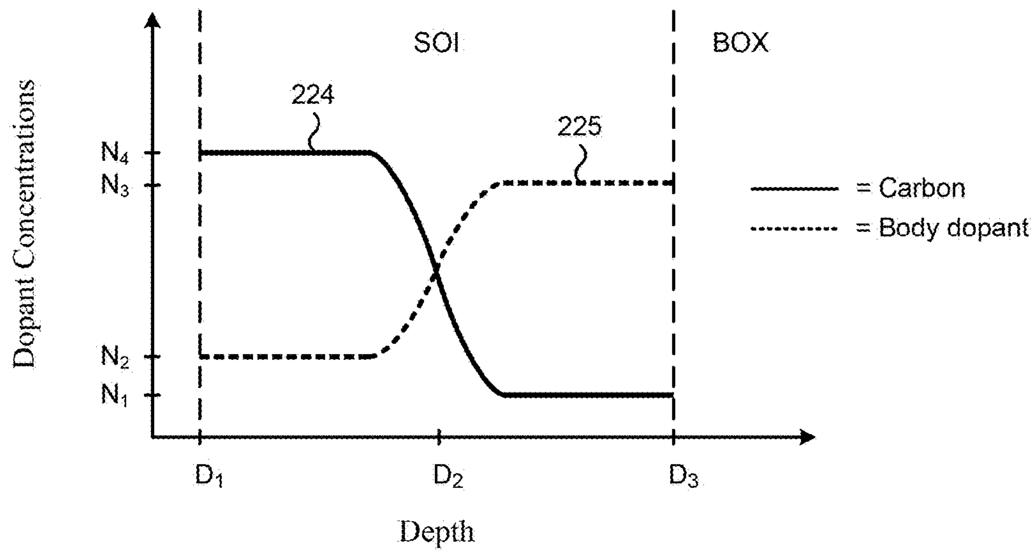

SOI
BOX
224
225
Dopant Concentrations
N_4
N_3
N_2
N_1
= Carbon
= Body dopant
D_1
D_2
D_3
Depth

FIG. 6

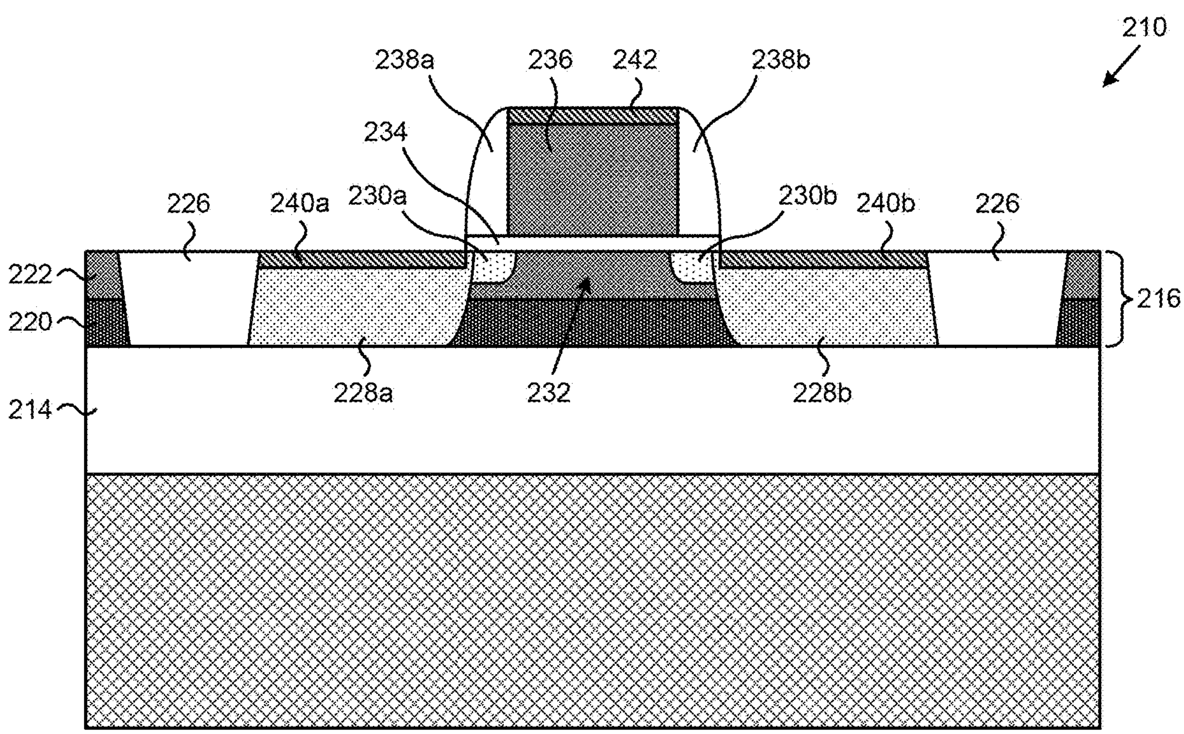

210
238a
236
242
238b
234
226
240a
230a
230b
240b
226
222
220
216
214
228a
232
228b

FIG. 7

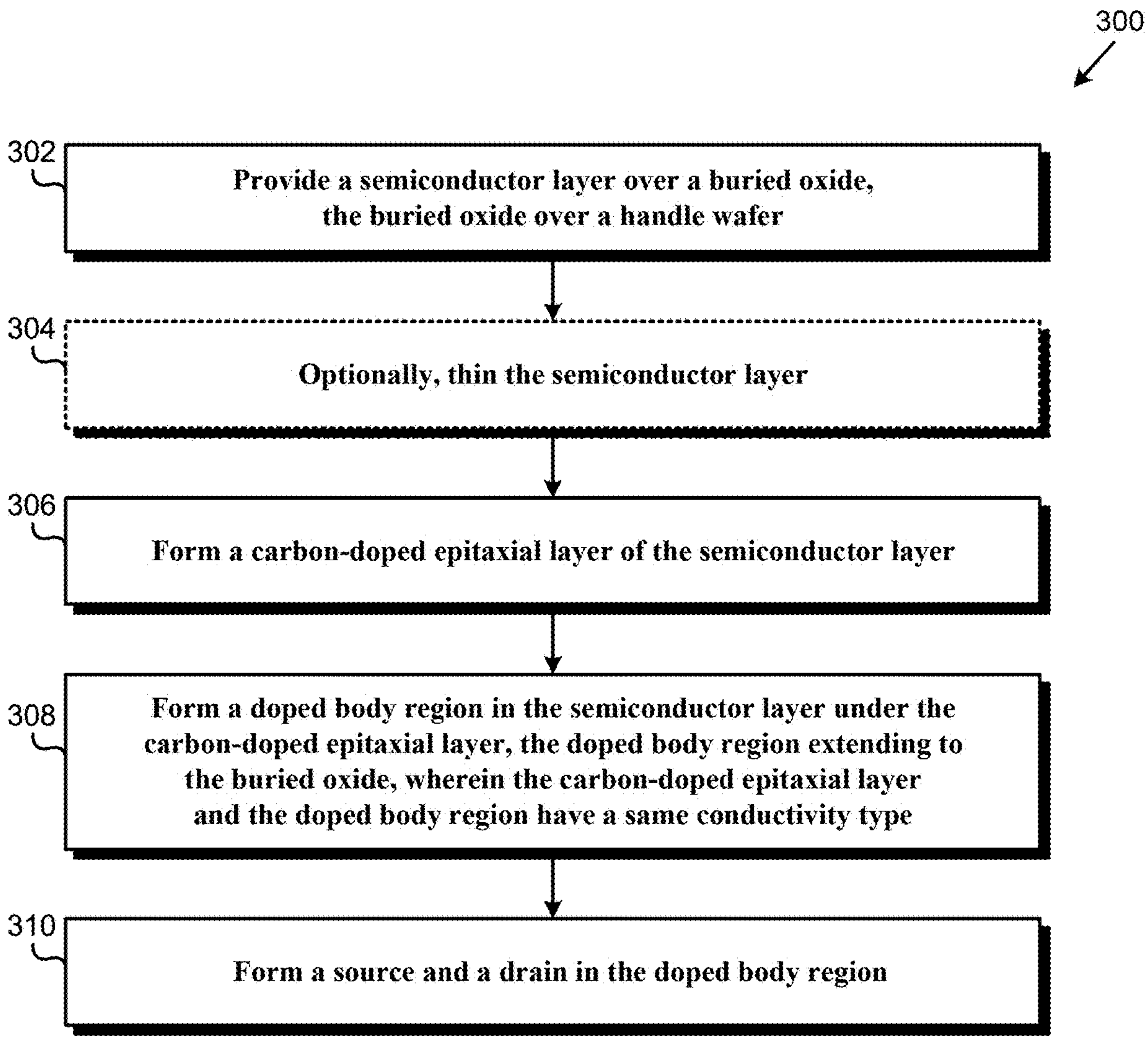

300
302
Provide a semiconductor layer over a buried oxide, the buried oxide over a handle wafer
304
Optionally, thin the semiconductor layer
306
Form a carbon-doped epitaxial layer of the semiconductor layer
308
Form a doped body region in the semiconductor layer under the carbon-doped epitaxial layer, the doped body region extending to the buried oxide, wherein the carbon-doped epitaxial layer and the doped body region have a same conductivity type
310
Form a source and a drain in the doped body region

FIG. 8

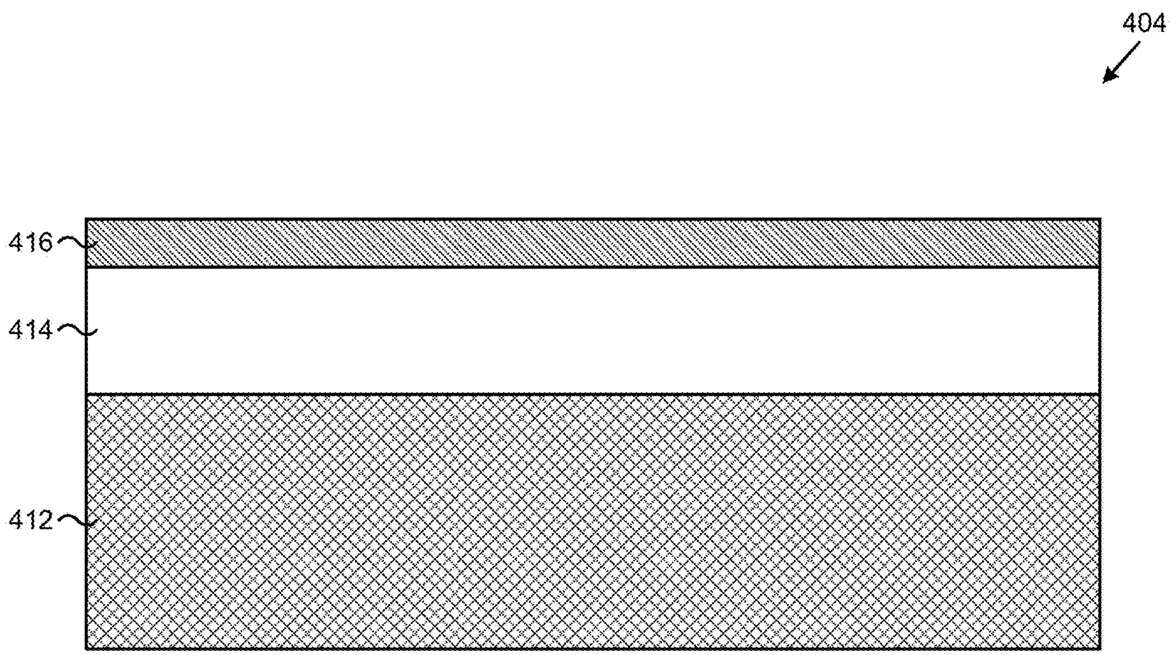


FIG. 9

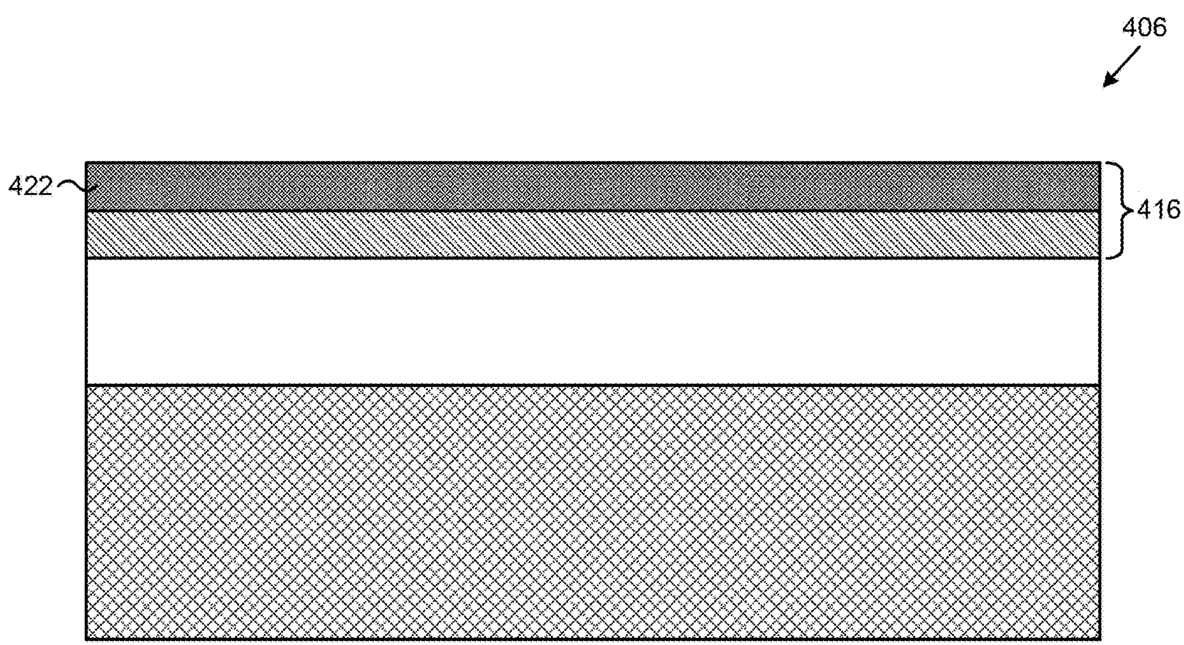


FIG. 10

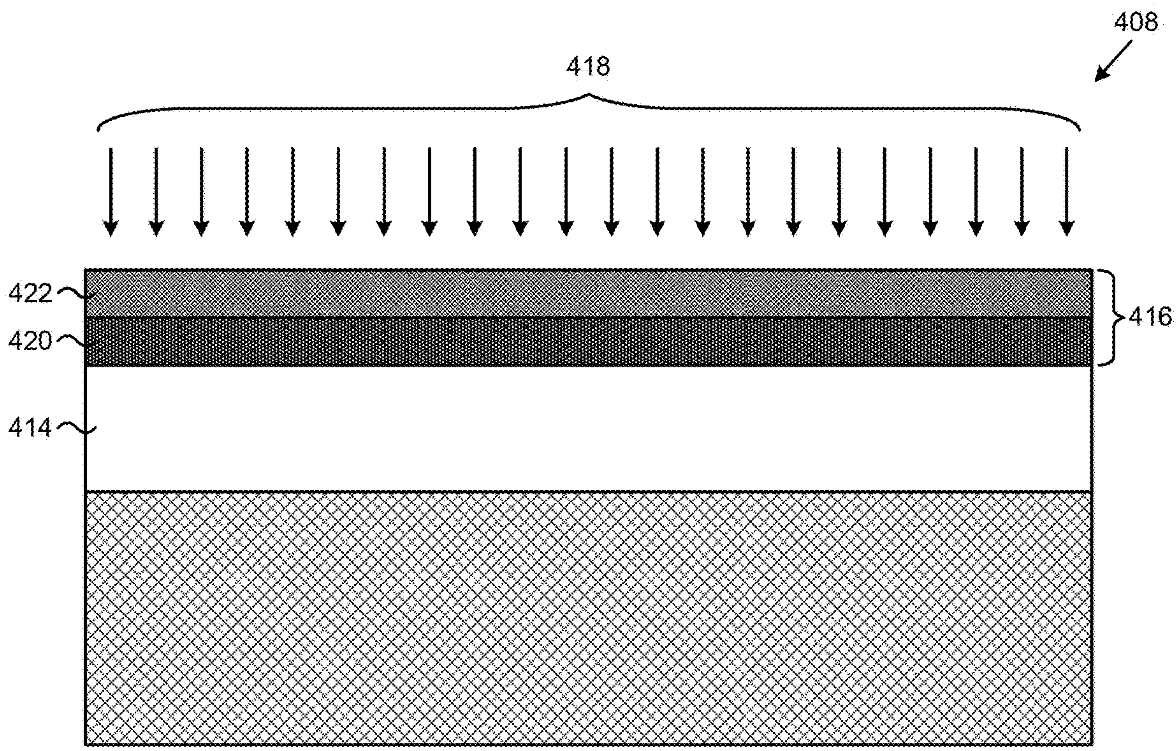

408
418
422
420
414
416

FIG. 11

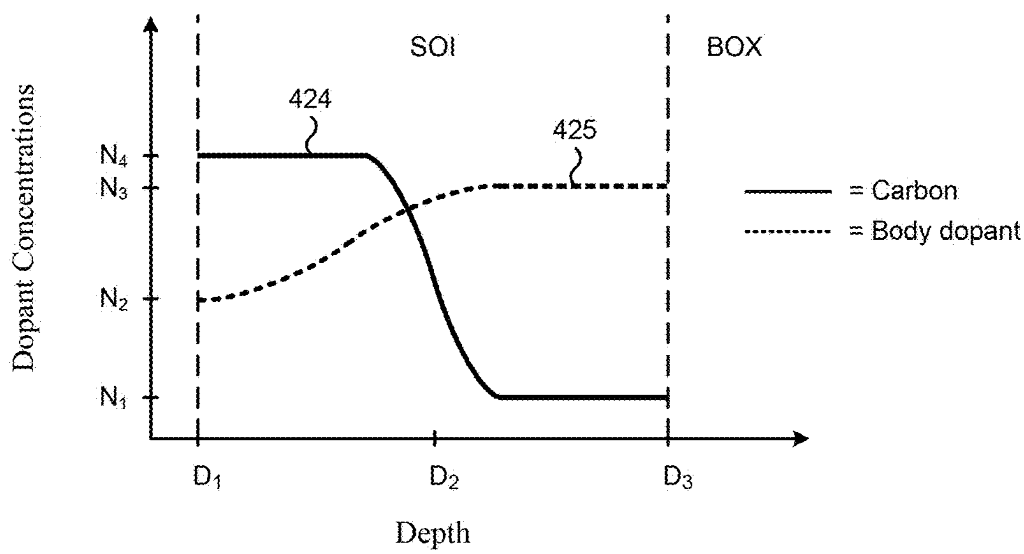

SOI
BOX
424
425
Dopant Concentrations
N4
N3
N2
N1
D1
D2
D3
Depth
= Carbon
= Body dopant

FIG. 12

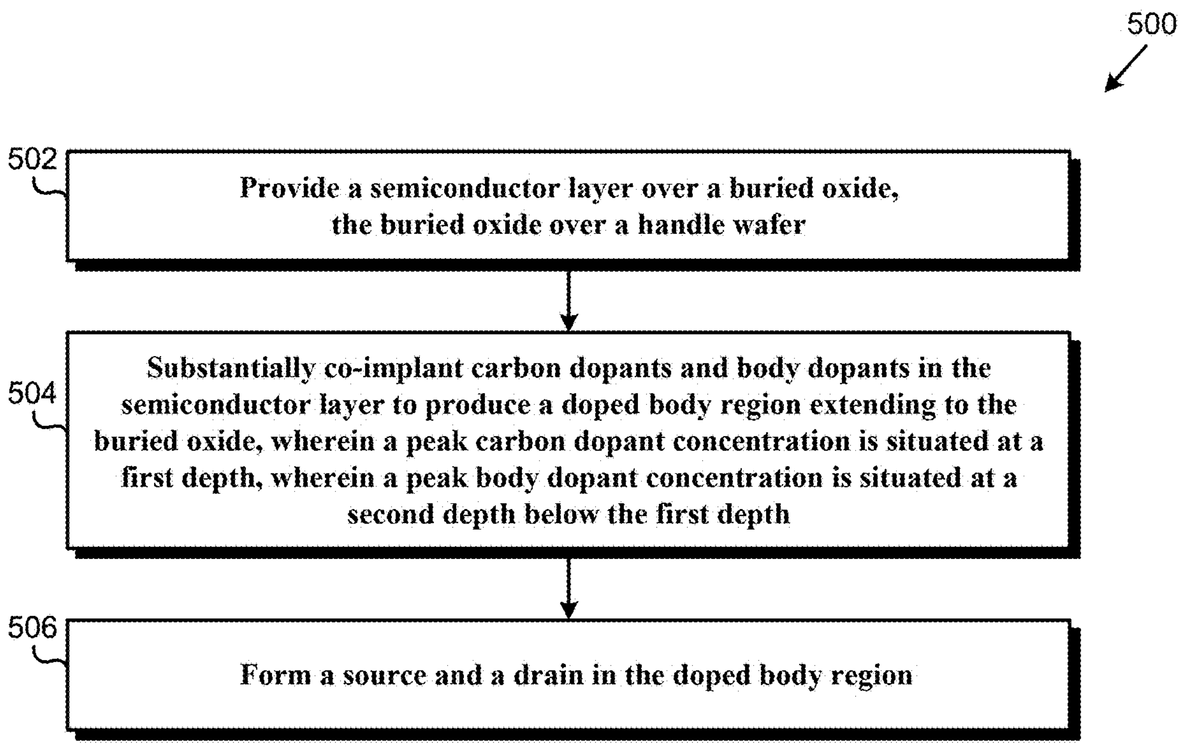

500
502
Provide a semiconductor layer over a buried oxide, the buried oxide over a handle wafer
504
Substantially co-implant carbon dopants and body dopants in the semiconductor layer to produce a doped body region extending to the buried oxide, wherein a peak carbon dopant concentration is situated at a first depth, wherein a peak body dopant concentration is situated at a second depth below the first depth
506
Form a source and a drain in the doped body region

FIG. 13

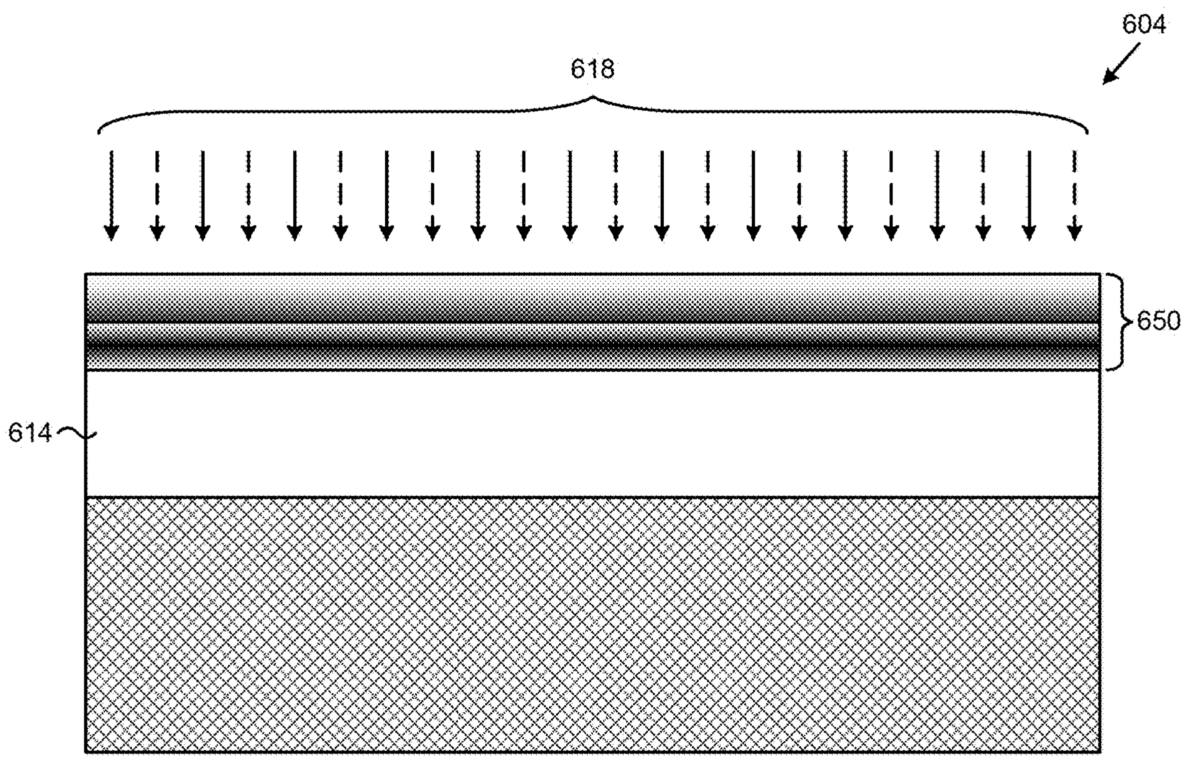

604
618
650
614

FIG. 14

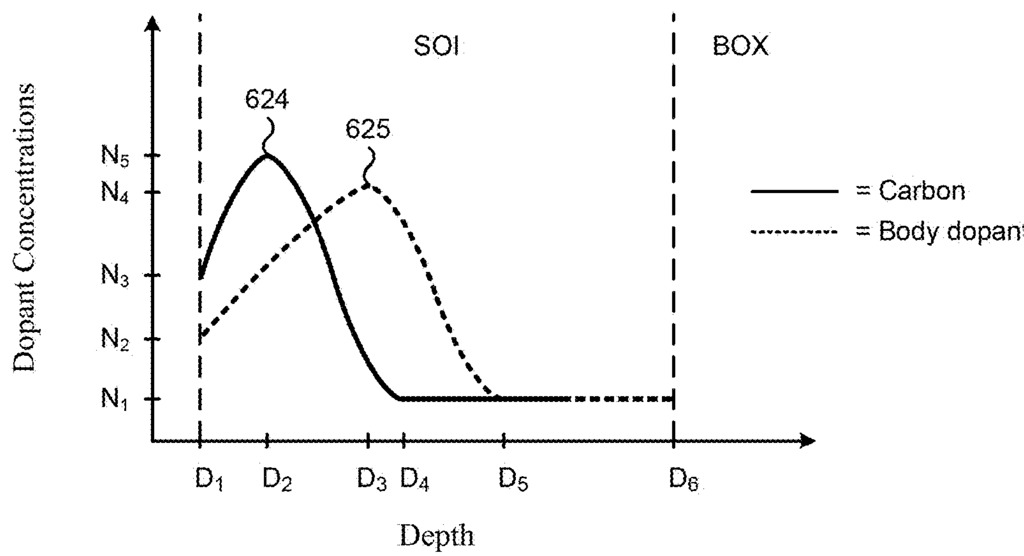

SOI
BOX
624
625
Dopant Concentrations
N5
N4
N3
N2
N1
= Carbon
= Body dopant
D1
D2
D3
D4
D5
D6
Depth

FIG. 15

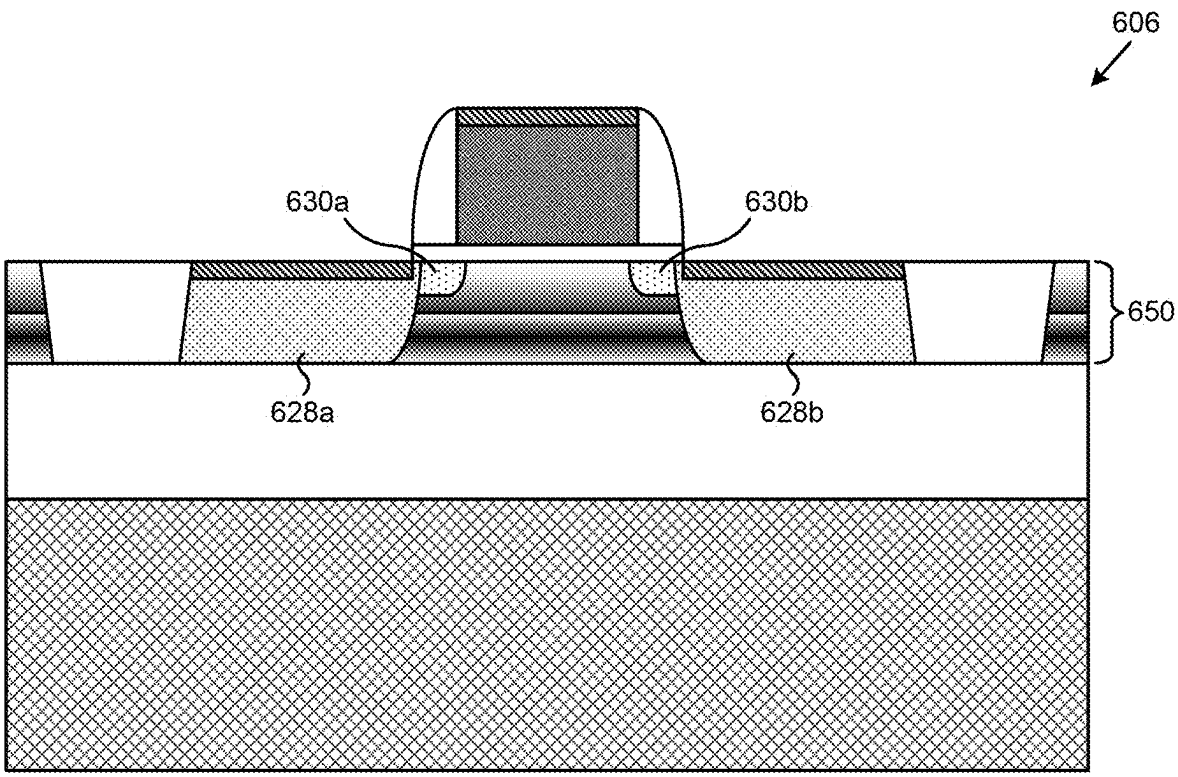


FIG. 16

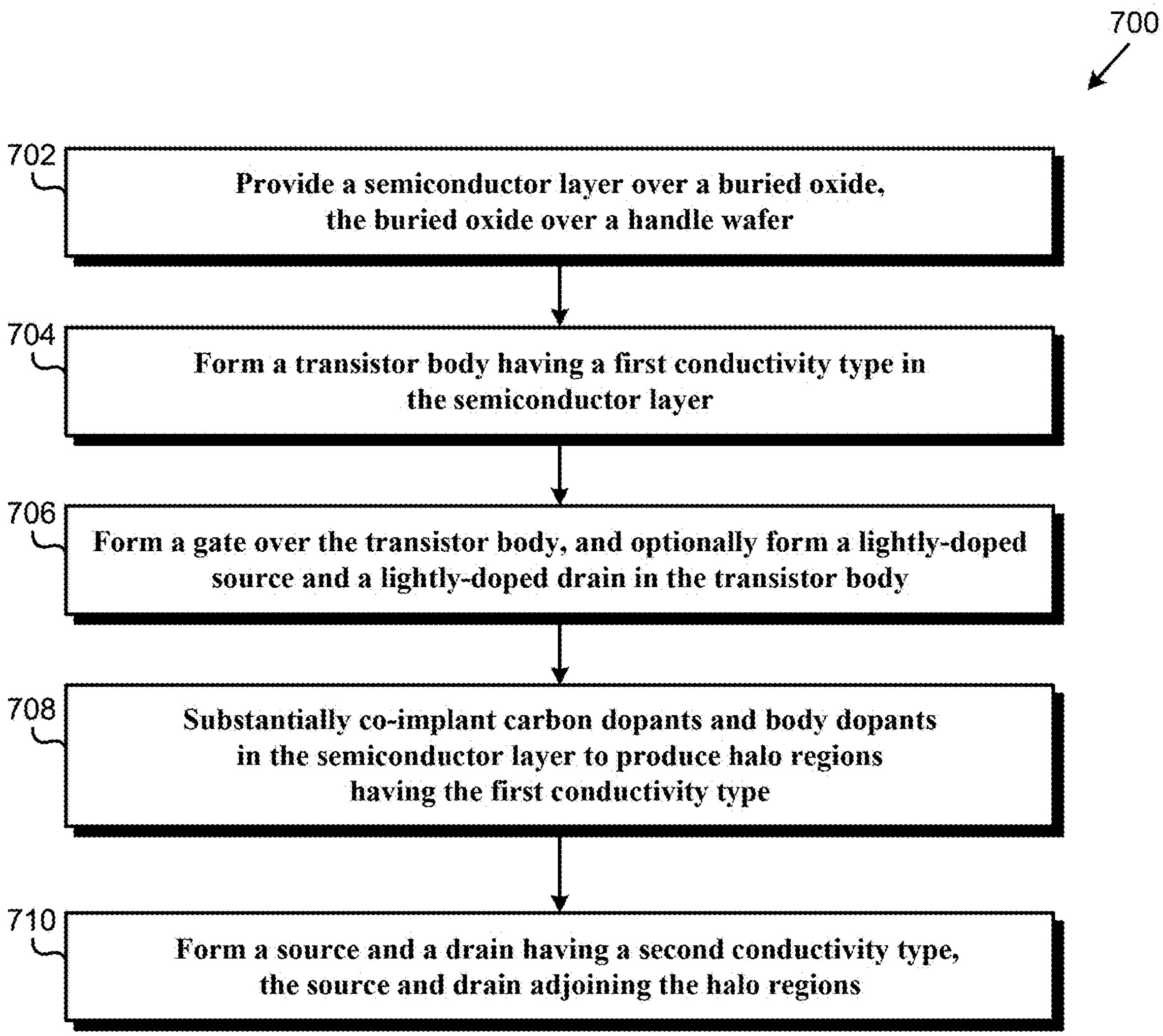

700
702
Provide a semiconductor layer over a buried oxide, the buried oxide over a handle wafer
704
Form a transistor body having a first conductivity type in the semiconductor layer
706
Form a gate over the transistor body, and optionally form a lightly-doped source and a lightly-doped drain in the transistor body
708
Substantially co-implant carbon dopants and body dopants in the semiconductor layer to produce halo regions having the first conductivity type
710
Form a source and a drain having a second conductivity type, the source and drain adjoining the halo regions

FIG. 17

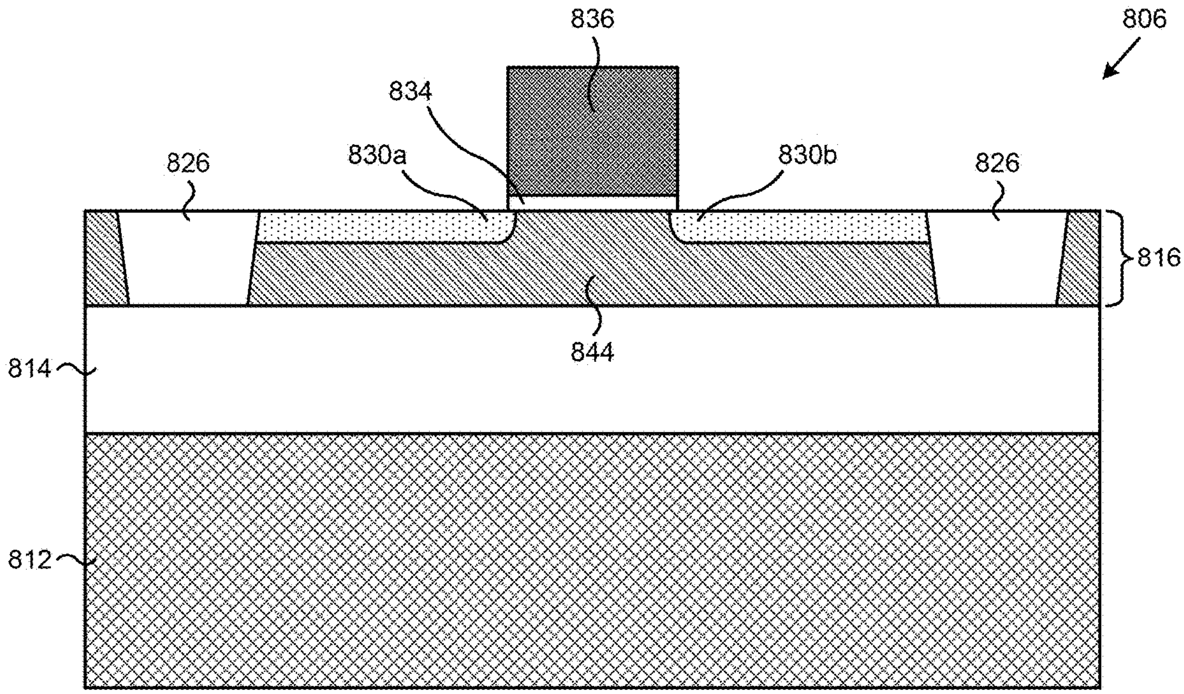

836
806
834
826
830a
830b
826
816
844
814
812

FIG. 18

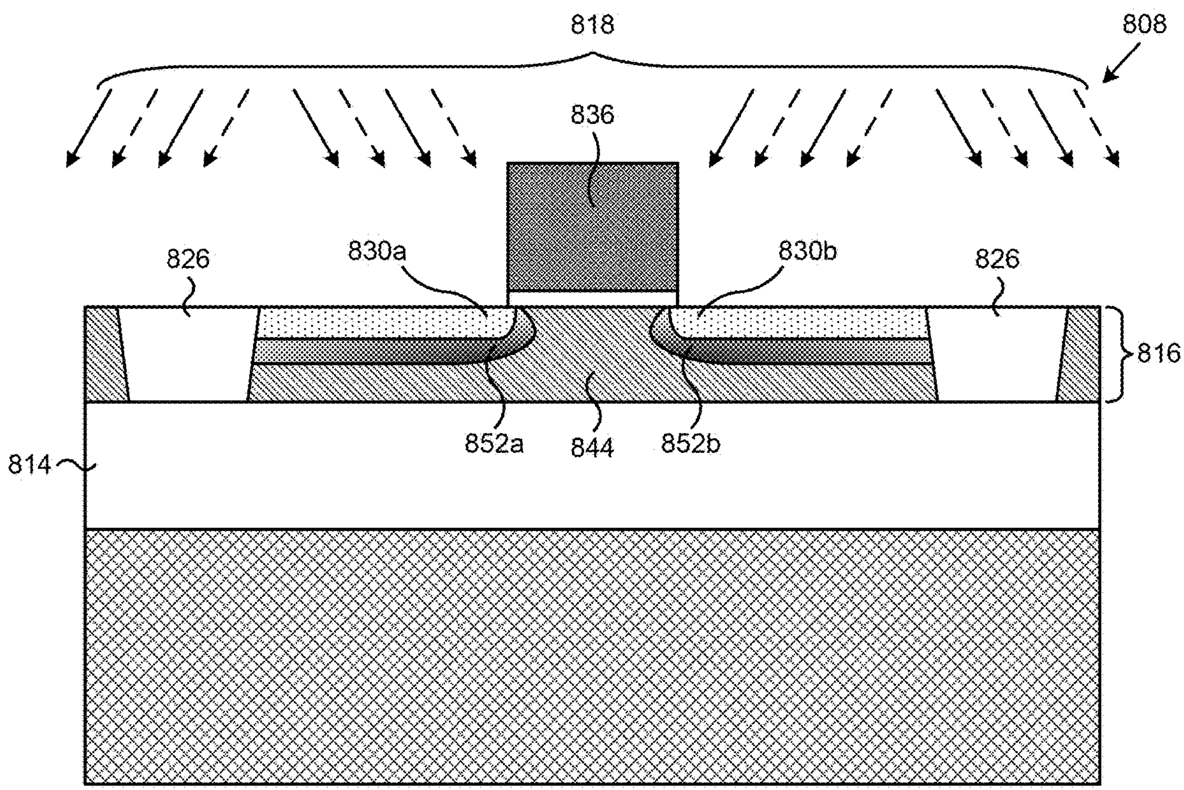

818
808
836
826
830a
830b
826
816
852a
844
852b
814

FIG. 19

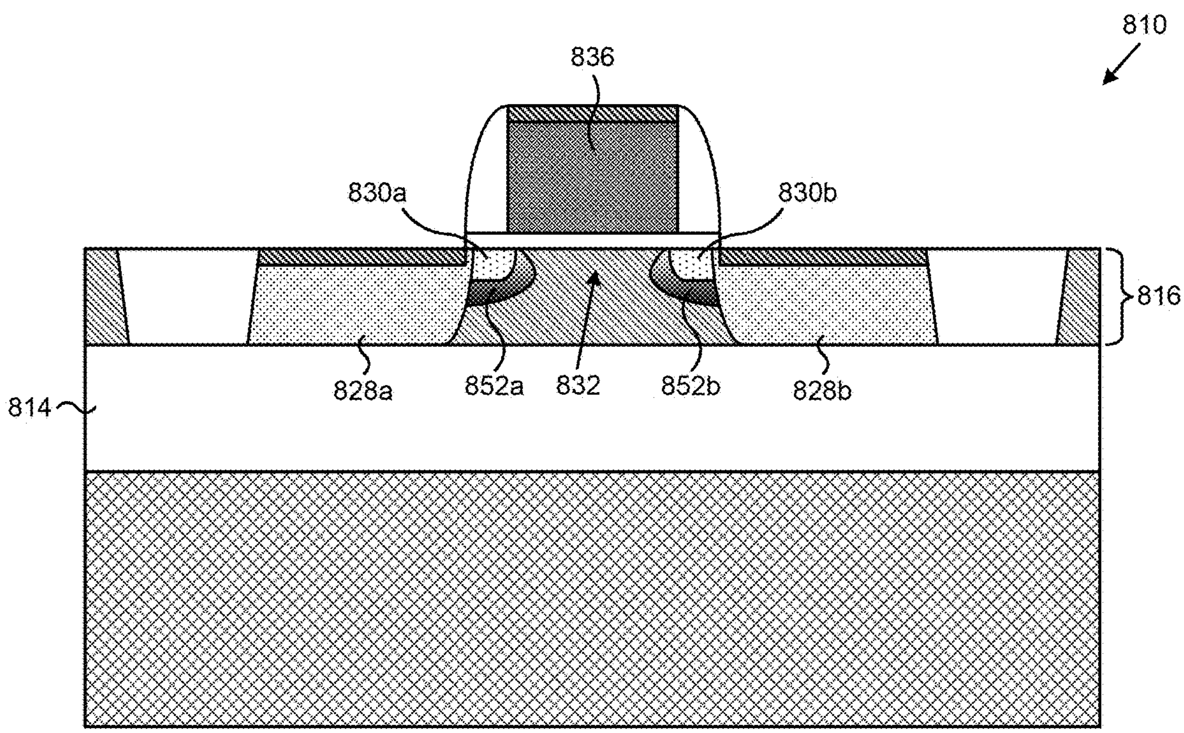


FIG. 20

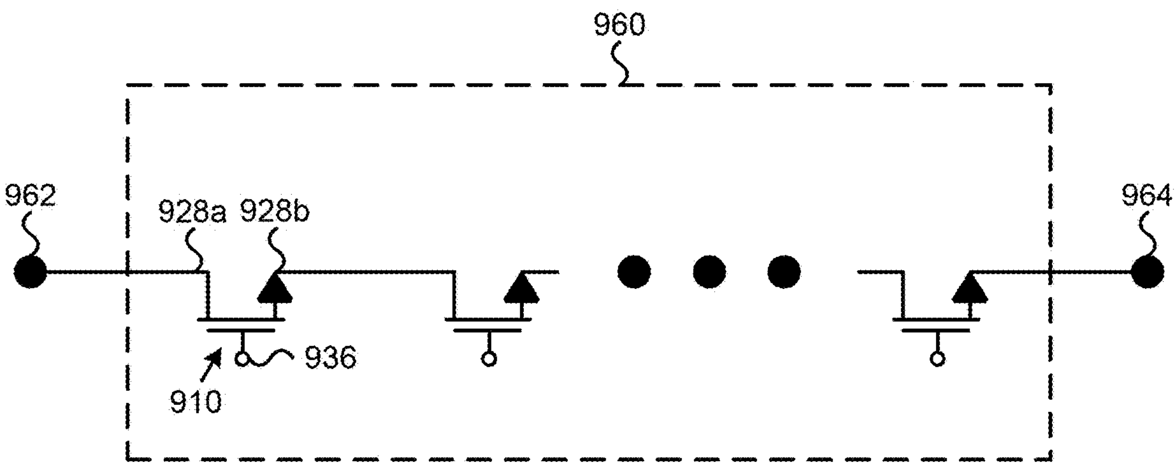


FIG. 21

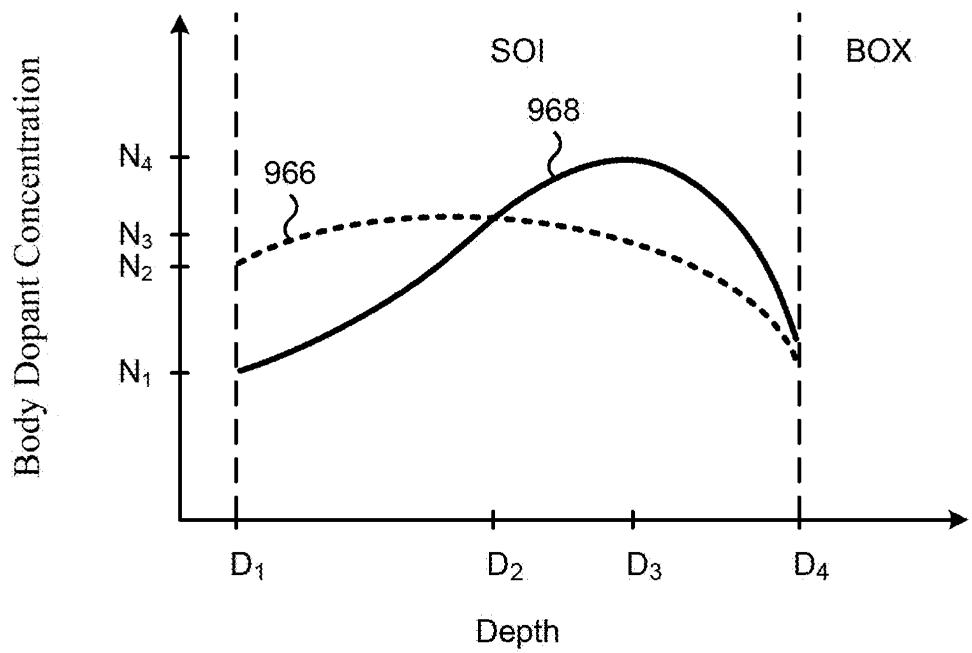

Body Dopant Concentration
SOI
BOX
968
966
N4
N3
N2
N1
D1
D2
D3
D4
Depth

FIG. 22

METHOD OF FABRICATING SOI DEVICE WITH CARBON IN BODY REGIONS

CLAIMS OF PRIORITY

[0001] The present application is a continuation-in-part of and claims the benefit of and priority to application Ser. No. 17/735,450 filed on May 3, 2022, titled "SOI Structures Including an Indium Retrograde P-Well for Improved RF-SOI Switches." The entire content of the above-identified application is hereby incorporated fully by reference into the present application.

BACKGROUND

[0002] Radio frequency (RF) switches are commonly utilized in wireless communication devices (e.g., smart phones) to route signals through transmit and receive paths, for example between the device's processing circuitry and the device's antenna. RF transistors, such as field effect transistor (FET) type RF transistors, can be arranged in a stack in order to improve RF power handling of RF switches. RF transistors can also be formed in semiconductor-on-insulator (SOI) substrates in order to provide insulation and reduce RF noise.

[0003] However, SOI substrates can be a source of other effects which degrade RF performance, such as punch-through effects, particularly when RF-SOI transistors are scaled down to have short channels. The product of ON-state resistance and OFF-state capacitance ($R_{ON}\times C_{OFF}$) is a figure of merit RF-SOI transistor designs strive to minimize in order to improve RF performance. During fabrication of SOI devices, especially during thermal processing, body dopants in a semiconductor layer can diffuse, thereby undermining conventional techniques for reducing the product $R_{ON}\times C_{OFF}$. Accordingly, fabricating RF-SOI transistors without significant RF performance tradeoffs, such as reduced breakdown voltage (V_{BD}) and reduced RF power handling, becomes difficult and complex.

[0004] Thus, there is a need in the art for RF-SOI transistors that have improved performance parameters with fewer tradeoffs.

SUMMARY

[0005] The present disclosure is directed to SOI structures with carbon in body regions for improved RF-SOI switches, substantially as shown in and/or described in connection with at least one of the figures, and as set forth in the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** illustrates a flowchart of an exemplary method for fabricating a semiconductor-on-insulator (SOI) structure according to one implementation of the present application.

[0007] FIG. **2** illustrates an SOI structure processed in accordance with the flowchart of FIG. **1** according to one implementation of the present application.

[0008] FIG. **3** illustrates an SOI structure processed in accordance with the flowchart of FIG. **1** according to one implementation of the present application.

[0009] FIG. **4** illustrates an SOI structure processed in accordance with the flowchart of FIG. **1** according to one implementation of the present application.

[0010] FIG. **5** illustrates an SOI structure processed in accordance with the flowchart of FIG. **1** according to one implementation of the present application.

[0011] FIG. **6** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application.

[0012] FIG. **7** illustrates an SOI structure processed in accordance with the flowchart of FIG. **1** according to one implementation of the present application.

[0013] FIG. **8** illustrates a flowchart of an exemplary method for fabricating an SOI structure according to one implementation of the present application.

[0014] FIG. **9** illustrates an SOI structure processed in accordance with the flowchart of FIG. **8** according to one implementation of the present application.

[0015] FIG. **10** illustrates an SOI structure processed in accordance with the flowchart of FIG. **8** according to one implementation of the present application.

[0016] FIG. **11** illustrates an SOI structure processed in accordance with the flowchart of FIG. **8** according to one implementation of the present application.

[0017] FIG. **12** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application.

[0018] FIG. **13** illustrates a flowchart of an exemplary method for fabricating an SOI structure according to one implementation of the present application.

[0019] FIG. **14** illustrates an SOI structure processed in accordance with the flowchart of FIG. **13** according to one implementation of the present application.

[0020] FIG. **15** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application.

[0021] FIG. **16** illustrates an SOI structure processed in accordance with the flowchart of FIG. **13** according to one implementation of the present application.

[0022] FIG. **17** illustrates a flowchart of an exemplary method for fabricating an SOI device according to one implementation of the present application.

[0023] FIG. **18** illustrates an SOI structure processed in accordance with the flowchart of FIG. **17** according to one implementation of the present application.

[0024] FIG. **19** illustrates an SOI structure processed in accordance with the flowchart of FIG. **17** according to one implementation of the present application.

[0025] FIG. **20** illustrates an SOI device processed in accordance with the flowchart of FIG. **17** according to one implementation of the present application.

[0026] FIG. **21** illustrates a portion of a circuit including a radio frequency (RF) switch employing stacked SOI transistors according to one implementation of the present application.

[0027] FIG. **22** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application.

DETAILED DESCRIPTION

[0028] The following description contains specific information pertaining to implementations in the present disclosure. The drawings in the present application and their accompanying detailed description are directed to merely exemplary implementations. Unless noted otherwise, like or corresponding elements among the figures may be indicated by like or corresponding reference numerals. Moreover, the drawings and illustrations in the present application are generally not to scale, and are not intended to correspond to actual relative dimensions.

[0029] FIG. **1** illustrates a flowchart of an exemplary method for manufacturing a semiconductor-on-insulator (SOI) structure according to one implementation of the present application. Structures shown in FIGS. **2** through **5** and **7** illustrate the results of performing actions **102** through **110** shown in flowchart **100** of FIG. **1**. For example, FIG. **2** shows an SOI structure after performing action **102** in FIG. **1**, FIG. **3** shows an SOI structure after performing action **104** in FIG. **1**, and so forth.

[0030] Actions **102** through **110** shown in flowchart **100** of FIG. **1** are sufficient to describe one implementation of the present inventive concepts. Other implementations of the present inventive concepts may utilize actions different from those shown in the flowchart of FIG. **1**. Certain details and features have been left out of the flowchart that are apparent to a person of ordinary skill in the art. For example, an action may consist of one or more sub-actions or may involve specialized equipment or materials, as known in the art. Moreover, some actions, such as masking and cleaning actions, are omitted so as not to distract from the illustrated actions.

[0031] FIG. **2** illustrates an SOI structure processed in accordance with action **102** in the flowchart of FIG. **1** according to one implementation of the present application. As shown in FIG. **2**, SOI structure **202** is provided. SOI structure **202** is an SOI substrate including handle wafer **212**, buried oxide (BOX) **214**, and semiconductor layer **216**.

[0032] In providing SOI structure **202**, a bonded and etch back SOI (BESOI) process can be used, as known in the art. In a BESOI process, handle wafer **212**, BOX **214**, and semiconductor layer **216** together form an SOI substrate. Alternatively, as also known in the art, a SIMOX process (separation by implantation of oxygen process) or a "smart cut" process can also be used for providing SOI structure **202**. In a SIMOX process, handle wafer **212** can be a bulk silicon support wafer (which for ease of reference, may still be referred to as a "handle wafer" in the present application). Similar to a BESOI process, in both SIMOX and smart cut processes, handle wafer **212**, BOX **214**, and semiconductor layer **216** together form an SOI substrate.

[0033] In one implementation, handle wafer **212** is undoped bulk silicon. In various implementations, handle wafer **212** can comprise germanium, group III-V material, or any other suitable handle material. In various implementations, handle wafer **212** has a thickness of approximately seven hundred microns (700 μm) or greater or less. In one implementation, a trap rich layer can be situated between handle wafer **212** and BOX **214**. In various implementations, BOX **214** typically comprises silicon dioxide (SiO_2), but it may also comprise silicon nitride (Si_XN_Y), or another insulator material. In various implementations, BOX **214** has a thickness of approximately one micron (1 μm) or greater or less. In one implementation, semiconductor layer **216** includes monocrystalline silicon. In various implementations, semiconductor layer **216** can comprise germanium, group III-V material, or any other semiconductor material. In various implementations, semiconductor layer **216** has a thickness of approximately three hundred nanometers (300 nm) or greater or less. Handle wafer **212**, BOX **214**, and semiconductor layer **216** can be provided together in SOI structure **202** as a pre-fabricated SOI substrate.

[0034] FIG. **3** illustrates an SOI structure processed in accordance with action **104** in the flowchart of FIG. **1** according to one implementation of the present application. As shown in FIG. **3**, in SOI structure **204**, semiconductor layer **216** is thinned.

[0035] Semiconductor layer **216** can be thinned using any means known in the art. In one implementation, semiconductor layer **216** can be thinned by oxidizing a portion of semiconductor layer **216** followed by a wet etching the oxidized portion, for example, using diluted hydrofluoric acid (DHF). In another implementation, semiconductor layer **216** can be thinned by grinding and/or chemical mechanical polishing (CMP). In various implementations, semiconductor layer **216** may be thinned to approximately half its original thickness, or greater or less.

[0036] FIG. **4** illustrates an SOI structure processed in accordance with action **106** in the flowchart of FIG. **1** according to one implementation of the present application. As shown in FIG. **4**, in SOI structure **206**, doped body region **220** is formed in semiconductor layer **216** and extending to BOX **214**.

[0037] Doped body region **220** is a region in semiconductor layer **216** comprising body dopants **218** and having a first conductivity type. In one implementation, doped body region **220** may consist of boron body dopants and have P-type conductivity. In another implementation, doped body region **220** may comprise a mixture of other P-type body dopants. In one implementation, doped body region **220** may be formed by ion implantation of body dopants **218** through a top surface of semiconductor layer **216**. In one implementation, body dopants **218** may comprise boron difluoride (BF_2) utilizing an implantation energy of approximately fifteen kiloelectronvolts (15 keV). In another implementation, body dopants **218** may comprise boron (B) utilizing an implantation energy of approximately two to six kiloelectron-volts (2-6 keV). In one implementation body dopants **218** may have an implant dosage of approximately $4.0*10_{12}/cm^3$ or greater or less. In one implementation, the ion implantation may be followed by rapid thermal processing (RTP) to heal implant damage and suppress transient enhanced diffusion (TED).

[0038] Notably, as shown in FIG. **4**, doped body region **220** is formed such that it extends to BOX **214**. That is, doped body region **220** is at the interface of semiconductor layer **216** with BOX **214**. In the implementation shown in FIG. **4**, doped body region **220** is formed along the entire width of SOI structure **206**, which may simplify fabrication. In various implementations, doped body region **220** may be formed in select regions along the width of SOI structure **206**, for example, by using a mask prior ion implantation. It is also noted that SOI structure **206** in FIG. **4** may be achieved by different fabrication actions and/or by different ordering of actions than those specifically described above. For example, a similar structure to SOI structure **206** may be achieved by implanting body dopants **218** in semiconductor layer **216** prior to thinning semiconductor layer **216**, that is, by switching actions **104** and **106** in the flowchart of FIG. **1**.

[0039] FIG. **5** illustrates an SOI structure processed in accordance with action **108** in the flowchart of FIG. **1** according to one implementation of the present application. As shown in FIG. **5**, in SOI structure **208**, carbon-doped epitaxial layer **222** of semiconductor layer **216** is formed over doped body region **220**.

[0040] Carbon-doped epitaxial layer **222** is an epitaxial layer of semiconductor layer **216** comprising both carbon dopants and body dopants having the first conductivity type. Continuing the above examples where semiconductor layer **216** includes monocrystalline silicon and doped body region **220** consists of P-type body dopants, carbon-doped epitaxial layer **222** may be epitaxial monocrystalline silicon having both carbon dopants and P-type body dopants. Carbon-doped epitaxial layer **222** has doped body region **220** thereunder. Carbon-doped epitaxial layer **222** and doped body region **220** have the same conductivity type. In one implementation, carbon-doped epitaxial layer **222** may have a lower concentration of body dopants compared to doped body region **220**.

[0041] Carbon-doped epitaxial layer **222** may be formed using any suitable techniques known in the art. In one implementation, carbon-doped epitaxial layer **222** of semiconductor layer **216** is formed over doped body region **220**, for example, using a low deposition-rate selective epitaxial growth (SEG). The dopants of carbon-doped epitaxial layer **222** may be introduced in situ while carbon-doped epitaxial layer **222** is formed. For example, carbon dopants may be introduced by flowing methylsilane at approximately one to fifty standard cubic centimeters per minute (1-50 SCCM), and body dopants may be introduced using chemical vapor deposition (CVD). In various implementations, carbon-doped epitaxial layer **222** may also be epitaxially grown using one of molecular beam epitaxy (MBE), atomic layer deposition (ALD), or low energy plasma-enhanced chemical vapor deposition (LEPECVD), and may be doped using one of diffusion or ion implantation. In various implementation, carbon-doped epitaxial layer **222** may have a thickness approximately equal to the thickness of semiconductor layer **216** that was removed during the thinning action **104** in the flowchart of FIG. **1**, such that semiconductor layer **216** in SOI structure **208** in FIG. **5** has approximately the same thickness as semiconductor layer **216** initially provided in SOI structure **202** in FIG. **2**.

[0042] FIG. **6** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application. In the graph, the y-axis represents the concentration of dopants in semiconductor layer **216** in FIG. **5**. The x-axis represents the depth in semiconductor layer **216** in FIG. **5**, where depth D_1 corresponds to a top surface of semiconductor layer **216**, depth D_2 corresponds to the bottom of carbon-doped epitaxial layer **222**, and depth D_3 corresponds to a bottom surface of semiconductor layer **216**. Trace **224** represents the concentration of carbon dopants in semiconductor layer **216**. Trace **225** represents the concentration of body dopants having the first conductivity type, such as boron dopants, in semiconductor layer **216**.

[0043] As shown by trace **224** in FIG. **6**, carbon dopants have a higher dopant concentration N_4 occurring from depth D_1 to approximately depth D_2, which generally corresponds to carbon-doped epitaxial layer **222**. From approximately depth D_2 to depth D_3, the dopant concentration of the carbon dopants falls significantly to lower dopant concentration N_1, which generally corresponds to doped body region **220**. Concentration N_1 may represent a negligible or substantially zero carbon concentration. As shown by trace **225**, body dopants have a lower dopant concentration N_2 occurring from depth D_1 to approximately depth D_2, which generally corresponds to carbon-doped epitaxial layer **222**. From approximately depth D_2 to depth D_3, the dopant concentration of the body dopants rises significantly to higher concentration N_3, which generally corresponds to doped body region **220**.

[0044] As shown by traces **224** and **225** in FIG. **6**, forming carbon-doped epitaxial layer **222** of semiconductor layer **216** over doped body region **220**, as shown in FIG. **5** and in accordance with action **108** in the flowchart of FIG. **1**, allows sharp dopant concentration transitions in semiconductor layer **216** near the bottom of carbon-doped epitaxial layer **222** (i.e., near depth D_2). Carbon dopants can be effectively localized to carbon-doped epitaxial layer **222**, with little or negligible carbon dopants thereunder. Also, semiconductor layer **216** can provide a retrograde doping profile with respect to body dopants having the first conductivity type. For example, a higher body dopant concentration can occur near the interface of semiconductor layer **216** with BOX **214** (i.e., between depths D_2 and D_3), while a lower body dopant concentration can occur in carbon-doped epitaxial layer **222** (i.e., between depths D_1 and D_2). In other implementations, the body dopant concentration between depths D_1 and D_2 may be higher than or similar to the body dopant concentration between depths D_2 and D_3. In various implementations, traces **224** and **225** may exhibit different patterns or slopes than shown in FIG. **6**.

[0045] FIG. **7** illustrates an SOI structure processed in accordance with action **110** in the flowchart of FIG. **1** according to one implementation of the present application. As shown in FIG. **7**, in SOI structure **210**, source **228***a* and drain **228***b* are formed in doped body region **220**. As also shown in FIG. **7**, in SOI structure **210**, additional components are also formed, including shallow trench isolations (STIs) **226**, lightly-doped source (LDS) **230***a*, lightly-doped drain (LDD) **230***b*, gate oxide **234**, gate **236**, dielectric spacers **238***a* and **238***b*, source/drain silicides **240***a* and **240***b*, and gate silicide **242**. SOI structure **210** in FIG. **7** includes a substantially completed SOI transistor in semiconductor layer **216**, and may also be referred to as SOI device **210** in the present application.

[0046] Shallow trench isolations (STIs) **226** are formed in doped body region **220**. More specifically, STIs **226** are formed in semiconductor layer **216**, extending through carbon-doped epitaxial layer **222** and into doped body region **220**. STIs **226** may comprise, for example, silicon dioxide (SiO_2) or another dielectric. STIs **226** may be formed in a manner known in the art. In the present implementation, STIs **226** are shown as extending to BOX **214**. In other implementations, STIs **226** may extend deeper or shallower in SOI structure **210**. In the present implementation, STIs **226** are formed after doped body region **220**, which helps ensure doped body region **220** extends uniformly to BOX **214**. For example, if STIs **226** were formed before doped body region **220**, STIs **226** could inhibit body dopants implanted through a top surface of semiconductor layer **216** from extending to BOX **214** in areas near the bottoms of STIs **226**. In various implementations, SOI structure **210** may include more or fewer STIs **226** than shown in FIG. **7**.

[0047] Source **228***a* and drain **228***b* are regions in semiconductor layer **216** having a second conductivity type opposite to the first conductivity type. Continuing the above examples where doped body region **220** and carbon-doped epitaxial layer **222** comprise P-type body dopants, source **228***a* and drain **228***b* may comprise N-type dopants, such as arsenic and/or phosphorus, for example, and may be formed in a manner known in the art. In SOI structure **210**, source **228***a* and drain **228***b* are situated between STIs **226**, in doped body region **220** and carbon-doped epitaxial layer **222**. Doped body region **220** extending to BOX **214** is situated between lower portions of source **228***a* and drain **228***b*. In the present implementation, source **228***a* and drain **228***b* extend to BOX **214**. In other implementations, source **228***a* and drain **228***b* may extend shallower in SOI structure **210**.

[0048] LDS **230***a* and LDD **230***b* are regions in semiconductor layer **216** having the second conductivity type and having a lower dopant concentration than source **228***a* and drain **228***b,* and may be formed in a manner known in the art. As shown in FIG. **7**, LDS **230***a* and LDD **230***b* are situated in carbon-doped epitaxial layer **222**, and do not extend to doped body region **220**. LDS **230***a* and LDD **230***b* are situated under dielectric spacers **238***a* and **238***b* and substantially aligned with sides of gate **236**. As known in the art, source **238***a* and LDS **230***a* together function as a source of SOI device **210**, while drain **238***b* and LDD **230***b* together function as a drain of SOI device **210**. Channel region **232** of SOI device **210** is between source **228***a* and drain **228***b,* and more particularly, between LDS **230***a* and LDD **230***b,* in carbon-doped epitaxial layer **222**.

[0049] Gate oxide **234** is situated on semiconductor layer **216** over LDS **230***a,* LDD **230***b,* and channel region **232**. Gate oxide **234** can comprise, for example, silicon dioxide (SiO_2) or another dielectric. Gate **236** is situated over gate oxide **234**. Gate **236** can comprise polycrystalline silicon (polysilicon) or a conductive metal. Dielectric spacers **238***a* and **238***b* are situated on sides of gate **236**. Dielectric spacers **238***a* and **238***b* can comprise, for example, silicon nitride. In the present implementation, source **228***a* and drain **228***b* are substantially aligned with dielectric spacers **268**.

[0050] Source silicide **240***a,* drain silicide **240***b,* and gate silicide **242** are situated on source **228***a,* drain **228***b,* and gate **236** respectively. Source silicide **240***a,* drain silicide **240***b,* and gate silicide **242** may be formed, for example, by sputtering a metal layer, such as a cobalt (Co) or nickel (Ni) layer, followed by performing a first thermal anneal (RTA) to cause the metal layer to react with source **228***a,* drain **228***b,* and gate **236** to form metal-rich source silicide **240***a,* drain silicide **240***b,* and gate silicide **242**. As known in the art, source silicide **240***a,* drain silicide **240***b,* and gate silicide **242** create highly conductive electrical connections to source **228***a,* drain **228***b,* and gate **236** respectively, and reduce corresponding contact resistances of SOI device **210**.

[0051] SOI device **210** in FIG. **7** represents a substantially completed device. However, SOI device **210** can also include fewer elements than shown in FIG. **7** and/or additional elements not shown in FIG. **7**. For example, a multilayer stack of metallizations and interlayer dielectrics can be situated over SOI device **210** to create connections to source silicide **240***a,* drain silicide **240***b,* and gate silicide **242**. As another example, passive devices, such as inductors and capacitors, can be integrated in an overlying multi-layer stack. As yet another example, additional transistors (not shown in FIG. **7**) can be situated in semiconductor layer **216**.

[0052] In SOI device **210**, carbon-doped epitaxial layer **222** of semiconductor layer **216** and doped body region **220** of semiconductor layer **216** have the same conductivity type and together function as a transistor body. Doped body region **220** is situated under carbon-doped epitaxial layer **222**. Carbon-doped epitaxial layer **222** inhibits body dopants of doped body region **220**, such as boron, from diffusing upwards during fabrication of SOI device **210** and during subsequent thermal processing, thereby maintaining a higher concentration of body dopants in doped body region **220** near the critical interface of semiconductor layer **216** with BOX **214**. Notably, doped body region **220** extends to this interface, and source **238***a* and drain **238***b* are situated in doped body region **220**. As a result, the higher concentration of body dopants in doped body region **220** advantageously inhibits short-channel effects that would otherwise occur at the interface of semiconductor layer **216** with BOX **214**, such as punch-through of source **228***a* and drain **228***b*.

[0053] Also notably, SOI device **212** includes LDS **230***a,* LDD **230***b,* and channel region **232** in carbon-doped epitaxial layer **222**. Where carbon-doped epitaxial layer **222** has a relatively low concentration of body dopants of the first conductivity type, channel region **232** experiences less current scattering and higher mobility when SOI device **210** is in an ON state. In other words, because of the retrograde doping profile between doped body region **220** and carbon-doped epitaxial layer **222**, SOI device **210** benefits from a lower ON-state resistance (R_{ON}). As another result, there is less counter-doping in regions where LDS **230***a* and LDD **230***b* are formed, and LDS **230***a* and LDD **230***b* have higher concentrations of dopants of the second conductivity type, thereby further lowering R_{ON}. In a similar manner, there is less counter-doping in upper portions of source **228***a* and drain **232***b,* and siliciding metal may react with portions having higher concentrations of dopants of the second conductivity type in order to form higher quality source silicide **240***a* and drain silicide **240***b,* thereby further lowering R_{ON}.

[0054] FIG. **8** illustrates a flowchart of an exemplary method for fabricating an SOI structure according to one implementation of the present application. Structures shown in FIGS. **9** through **11** illustrate the results of performing actions **302** through **308** shown in flowchart **300** of FIG. **8**. For example, FIG. **9** shows an SOI structure after performing actions **302** and **304** in FIG. **8**, FIG. **10** shows an SOI structure after performing action **306** in FIG. **8**, and so forth.

[0055] Actions **302** through **310** shown in flowchart **300** of FIG. **8** are sufficient to describe one implementation of the present inventive concepts. Other implementations of the present inventive concepts may utilize actions different from those shown in the flowchart of FIG. **8**. Certain details and features have been left out of the flowchart that are apparent to a person of ordinary skill in the art. For example, an action may consist of one or more sub-actions or may involve specialized equipment or materials, as known in the art. Moreover, some actions, such as masking and cleaning actions, are omitted so as not to distract from the illustrated actions.

[0056] FIG. **9** illustrates an SOI structure processed in accordance with actions **302** and **304** in the flowchart of FIG. **8** according to one implementation of the present application. As shown in FIG. **9**, SOI structure **404** including handle wafer **412**, BOX **414**, and semiconductor layer **416** has been provided, and semiconductor layer **416** has been thinned. SOI structure **404** in FIG. **8** generally corresponds to SOI structure **204** in FIG. **3**, and may have any implementations and advantages described above.

[0057] FIG. **10** illustrates an SOI structure processed in accordance with action **306** in the flowchart of FIG. **8** according to one implementation of the present application.

As shown in FIG. **10**, in SOI structure **406**, carbon-doped epitaxial layer **422** of semiconductor layer **416** is formed.

[0058] As described above, carbon-doped epitaxial layer **422** is an epitaxial layer of semiconductor layer **416** comprising both carbon dopants and body dopants having the first conductivity type. Referring back to FIG. **5**, carbon-doped epitaxial layer **222** in FIG. **5** has doped body region **220** thereunder when carbon-doped epitaxial layer **222** is formed, whereas carbon-doped epitaxial layer **422** in FIG. **10** does not have such a doped body region thereunder when carbon-doped epitaxial layer **422** is formed. Otherwise, carbon-doped epitaxial layer **422** in FIG. **10** generally corresponds to carbon-doped epitaxial layer **222** in FIG. **5** and may have any implementations and advantages described above.

[0059] FIG. **11** illustrates an SOI structure processed in accordance with action **308** in the flowchart of FIG. **8** according to one implementation of the present application. As shown in FIG. **11**, in SOI structure **408**, doped body region **420** is formed in semiconductor layer **416** under carbon-doped epitaxial layer **422** and extending to BOX **414**. Except for differences noted below, doped body region **420** in FIG. **11** generally corresponds to doped body region **220** in FIG. **4** and may have any implementations and advantages described above.

[0060] As described above, doped body region **420** is a region in semiconductor layer **416** comprising body dopants **418** and having a first conductivity type. Notably, doped body region **420** in FIG. **11** is formed through carbon-doped epitaxial layer **422**. Accordingly, formation of doped body region **420** in FIG. **11** may utilize a higher implantation energy and implant dosage compared to formation of doped body region **220** in FIG. **4**. In one implementation, body dopants **218** may comprise BF_2 having an implant dosage of approximately $5.0*10^{12}/cm^3$ to $4.0*10^{13}/cm^3$, and may utilize an implantation energy of approximately twenty five kiloelectronvolts (25 keV).

[0061] FIG. **12** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application. In the graph, the y-axis represents the concentration of dopants in semiconductor layer **416** in FIG. **11**. The x-axis represents the depth in semiconductor layer **416** in FIG. **11**, where depth D_1 corresponds to a top surface of semiconductor layer **416**, depth D_2 corresponds to the bottom of carbon-doped epitaxial layer **422**, and depth D_3 corresponds to a bottom surface of semiconductor layer **416**. Trace **424** represents the concentration of carbon dopants in semiconductor layer **416**. Trace **425** represents the concentration of body dopants having the first conductivity type, such as boron dopants, in semiconductor layer **416**.

[0062] As shown by trace **424** in FIG. **12**, carbon dopants have a higher dopant concentration N_4 occurring from depth D_1 to approximately depth D_2, which generally corresponds to carbon-doped epitaxial layer **422**. From approximately depth D_2 to depth D_3, the dopant concentration of the carbon dopants falls significantly to lower dopant concentration N_1, which generally corresponds to doped body region **420**. Concentration N_1 may represent a negligible or substantially zero carbon concentration. As shown by trace **425**, body dopants have a low dopant concentration N_2 occurring at depth D_1. From depth D_1 to approximately depth D_2, the dopant concentration of the body dopants rises gradually to higher dopant concentration N_3, which generally corresponds to carbon-doped epitaxial layer **422**. From approximately depth D_2 to depth D_3, the dopant concentration of the body dopants remains at higher concentration N_3, which generally corresponds to doped body region **420**.

[0063] As shown by traces **424** and **425** in FIG. **12**, forming doped body region **420** in semiconductor layer **416** under carbon-doped epitaxial layer **422**, as shown in FIG. **11** and in accordance with action **308** in the flowchart of FIG. **8**, allows a sharp carbon dopant concentration transition in semiconductor layer **416** near the bottom of carbon-doped epitaxial layer **422** (i.e., near depth D_2), while body dopants having the first conductivity type can exhibit a more gradual dopant concentration transition in semiconductor layer **416** and still provide a retrograde doping profile. For example, comparing trace **425** in FIG. **12** to trace **225** in FIG. **6**, trace **425** in FIG. **12** generally has higher body dopant concentration N_2 in carbon-doped epitaxial layer **422** (i.e., between depths D_1 and D_2), and generally has a more gradual transition to its peak body dopant concentration N_3 in doped body region **420** (i.e., between depths D_2 and D_3), due to body dopants **418** being formed through carbon-doped epitaxial layer **422**.

[0064] In various implementations, traces **424** and **425** may exhibit different patterns or slopes than shown in FIG. **12**. It is also noted that SOI structure **408** in FIG. **11** may be achieved by different fabrication actions and/or by different ordering of actions than those specifically described above. For example, a similar structure to SOI structure **408** may be achieved by replacing carbon-doped epitaxial layer **422** formed in FIG. **10** with an epitaxial layer having carbon dopants but devoid of body dopants, then implanting body dopants **418** as shown in FIG. **11**. In such example, carbon-doped epitaxial layer **422** and doped body region **420** may be formed substantially concurrently when body dopants **418** are implanted into semiconductor layer **416**.

[0065] Thus, SOI structure **408** in FIG. **11** manufactured according to the flowchart in FIG. **8** generally represents an alternative to SOI structure **208** in FIG. **5** manufactured according to the flowchart in FIG. **2**. Both SOI structures **208** and **408** enable sharp transitions in carbon dopant concentrations. SOI structure **208** may more readily enable sharp transitions in body dopant concentration, while SOI structure **408** may more readily enable gradual transitions in body dopant concentration and simplified processing.

[0066] Processing of SOI structure **408** in FIG. **11** can continue with action **310** in the flowchart in FIG. **8** by forming a source and a drain in doped body region **420**, and forming a transistor in semiconductor layer **416** in order to complete an SOI device. Except for differences noted above, the resulting SOI device may appear similar to SOI device **210** shown in FIG. **7**, and may include the additional components shown therein. The resulting SOI device may also have any implementations and advantages described above. In particular, as described above, carbon-doped epitaxial layer **422** inhibits body dopants of doped body region **420**, such as boron, from diffusing upwards during fabrication of the SOI device and during subsequent thermal processing.

[0067] FIG. **13** illustrates a flowchart of an exemplary method for fabricating an SOI structure according to one implementation of the present application. Structures shown in FIGS. **14** and **16** illustrate the results of performing actions **502** through **506** shown in flowchart **500** of FIG. **13**. For example, FIG. **14** shows an SOI structure after performing actions **502** and **504** in FIG. **13**, FIG. **16** shows an SOI structure after performing action **506** in FIG. **13**.

[0068] Actions **502** through **506** shown in flowchart **500** of FIG. **13** are sufficient to describe one implementation of the present inventive concepts. Other implementations of the present inventive concepts may utilize actions different from those shown in the flowchart of FIG. **13**. Certain details and features have been left out of the flowchart that are apparent to a person of ordinary skill in the art. For example, an action may consist of one or more sub-actions or may involve specialized equipment or materials, as known in the art. Moreover, some actions, such as masking and cleaning actions, are omitted so as not to distract from the illustrated actions.

[0069] Processing of an SOI structure according to the flowchart of FIG. **13** can begin with action **502** by providing an SOI structure including a handle wafer, BOX, and semiconductor layer. The provided SOI structure generally corresponds to SOI structure **202** in FIG. **2**, and may have any implementations and advantages described above.

[0070] FIG. **14** illustrates an SOI structure processed in accordance with action **504** in the flowchart of FIG. **13** according to one implementation of the present application. As shown in FIG. **14**, in SOI structure **604**, carbon dopants and body dopants, collectively referred to as dopants **618** in the present application, are substantially co-implanted in the semiconductor layer to produce doped body region **650** extending to BOX **614**. As used in the present application, "substantially co-implanting" refers to implanting carbon dopants and body dopants either concurrently or back-to-back without intermediate processing actions.

[0071] Doped body region **650** is a region of the semiconductor layer comprising both carbon dopants and body dopants **618** having the first conductivity type. In one implementation, doped body region **650** may comprise carbon dopants and boron body dopants, and may have P-type conductivity. The implantation energies and implant dosages of dopants **618** are configured such that a peak carbon dopant concentration is situated at a first depth in doped body region **650**, and a peak body dopant concentration is situated at a second depth in doped body region **650** below the first depth, as described below. In one implementation, carbon dopants of dopants **618** may have an implant dosage of approximately $1.0*10^{15}/cm^3$, and may utilize an implantation energy of approximately fifteen kiloelectronvolts (25 keV), while body dopants of dopants **618** may have an implant dosage of approximately $4.0*10^{12}/cm^3$ to $1.0*10^{13}/cm^3$, and may utilize an implantation energy of approximately twelve to twenty kiloelectronvolts (12-20 keV).

[0072] In the implementation shown in FIG. **14**, doped body region **650** is formed along the entire width of SOI structure **604**, such that doped body region **650** is substantially coextensive with the semiconductor layer, which may simplify fabrication. In various implementations, doped body region **650** may be formed in select regions of the semiconductor layer along the width of SOI structure **604**, for example, by using a mask prior ion implantation.

[0073] FIG. **15** illustrates an exemplary graph of dopant concentrations versus depth according to one implementation of the present application. In the graph, the y-axis represents the concentration of dopants in doped body region **650** in FIG. **14**. The x-axis represents the depth in doped body region **650** in FIG. **14**, where depth D_1 corresponds to a top surface of the semiconductor layer, and depth D_6 corresponds to a bottom surface of the semiconductor layer. Trace **624** represents the concentration of carbon dopants in doped body region **650**. Trace **625** represents the concentration of body dopants having the first conductivity type, such as boron dopants, in the doped body region **650**.

[0074] As shown by trace **624** in FIG. **15**, carbon dopants have an initial carbon dopant concentration N_3 at depth D_1, which then rises to a peak carbon dopant concentration N_5 at depth D_2, which then falls to low carbon dopant concentration N_1 at approximately depth D_4. Concentration N_1 may represent a negligible or substantially zero carbon concentration. As shown by trace **625**, body dopants have an initial body dopant concentration N_2 at depth D_1, which then rises to a peak body dopant concentration N_4 at depth D_3, which then falls to low body dopant concentration N_1 at approximately depth D_5.

[0075] As shown by traces **624** and **625** in FIG. **15**, co-implanting carbon dopants and body dopants **618** as shown in FIG. **14** and in accordance with action **504** in the flowchart of FIG. **13**, produces doped body region **650** having a peak carbon dopant concentration at a higher depth D_2 and having a peak body dopant concentration at a lower depth D_3. As a result, the carbon dopants in body region **650** effectively inhibit body dopants, such as boron, in body region **650** from diffusing upwards during fabrication of SOI structure **604** and during subsequent processing, as described above. Thus, SOI structure **604** in FIG. **14** manufactured according to the flowchart in FIG. **13** generally represents an alternative to SOI structure **208** in FIG. **5** manufactured according to the flowchart in FIG. **2** and SOI structure **408** in FIG. **11** manufactured according to the flowchart in FIG. **8**.

[0076] Notably, because SOI structure **604** in FIG. **14** utilizes a carbon implant rather than a carbon epitaxy, less carbon dopants are generally concentrated near the top surface of the semiconductor layer. For example, comparing trace **624** in FIG. **15** to trace **224** in FIG. **6** or trace **424** in FIG. **12**, trace **624** in FIG. **15** generally has lower carbon dopant concentration near the top surface of the semiconductor layer (i.e., near depth D_1). As a result, a subsequently formed gate oxide (not shown in FIG. **14**) can be formed on the semiconductor layer with increased reliability. Also because SOI structure **604** utilizes a carbon implant rather than a carbon epitaxy, SOI structure **604** can more readily enable a range of transitions in carbon dopant concentration, gradual or sharp, depending on implant energy and dopant concentration. In various implementations, traces **624** and **625** may exhibit different patterns or slopes than shown in FIG. **15**.

[0077] FIG. **16** illustrates an SOI structure processed in accordance with action **506** in the flowchart of FIG. **13** according to one implementation of the present application. As shown in FIG. **16**, a transistor including source **628*a*** and drain **628*b*** is formed in doped body region **650**, in order to complete SOI device **606**.

[0078] In one implementation source **628*a*** and drain **628*b*** extend to a depth below the peak carbon dopant concentration in doped body region **650** (i.e., below depth D_2 in FIG. **15**). In one implementation LDS **630*a*** and LDD **630*b*** extend to a depth above the peak carbon dopant concentration in doped body region **650** (i.e., above depth D_2 in FIG. **15**). Notably, fabrication of SOI device **606** does not require semiconductor layer thinning or carbon epitaxy. Instead, carbon dopants are substantially co-implanted with body dopants. As described above, carbon dopants in body region **650** inhibit body dopants, such as boron, in body region **650** from diffusing upwards during fabrication of SOI device **606** and during subsequent thermal processing. Except for differences noted above, SOI device **606** in FIG. **16** may generally correspond to SOI device **210** in FIG. **7**, and may have any implementations and advantages described above.

[0079] FIG. **17** illustrates a flowchart of an exemplary method for fabricating an SOI device according to one implementation of the present application. Structures shown in FIGS. **18** through **20** illustrate the results of performing actions **702** through **710** shown in flowchart **700** of FIG. **17**. For example, FIG. **18** shows an SOI structure after performing actions **702**, **704**, and **706** in FIG. **17**, FIG. **19** shows an SOI structure after performing action **708** in FIG. **17**, and so forth.

[0080] Actions **702** through **710** shown in flowchart **700** of FIG. **17** are sufficient to describe one implementation of the present inventive concepts. Other implementations of the present inventive concepts may utilize actions different from those shown in the flowchart of FIG. **17**. Certain details and features have been left out of the flowchart that are apparent to a person of ordinary skill in the art. For example, an action may consist of one or more sub-actions or may involve specialized equipment or materials, as known in the art. Moreover, some actions, such as masking and cleaning actions, are omitted so as not to distract from the illustrated actions.

[0081] FIG. **18** illustrates an SOI structure processed in accordance with actions **702**, **704**, and **706** in the flowchart of FIG. **17** according to one implementation of the present application. As shown in FIG. **18**, in SOI structure **806**, an SOI substrate including handle wafer **812** over BOX **814** over semiconductor layer **816** has been provided. The provided SOI substrate generally corresponds to SOI structure **202** in FIG. **2**, and may have any implementations and advantages described above.

[0082] Transistor body **844** having a first conductivity type has been formed in the semiconductor layer. Transistor body **844** may be formed by ion implantation of body dopants through a top surface of semiconductor layer **816**. In the implementation shown in FIG. **18**, transistor body **844** is formed along the entire width of SOI structure **806**, which may simplify fabrication. In various implementations, transistor body **844** may be formed in select regions along the width of SOI structure **806**, for example, by using a mask prior ion implantation.

[0083] In SOI structure **806**, gate **836** has been formed over transistor body **844**. Gate **836** is situated between STIs **826** and has gate oxide **834** thereunder. Gate **836**, STIs **826**, and gate oxide **834** in FIG. **18** generally correspond to gate **236**, STIs **226**, and gate oxide **234** in FIG. **2**, and may have any implementations and advantages described above. Also in SOI structure **806**, LDS **830*a*** and LDD **830*b*** have been formed in transistor body **844**. LDS **830*a*** and LDD **830*b*** have the second conductivity type and are substantially aligned with sides of gate **836**. LDS **830*a*** and LDD **830*b*** in FIG. **18** generally correspond to LDS **230*a*** and LDD **230*b*** in FIG. **2**, and may have any implementations and advantages described above.

[0084] FIG. **19** illustrates an SOI structure processed in accordance with action **708** in the flowchart of FIG. **17** according to one implementation of the present application. As shown in FIG. **19**, in SOI structure **808**, carbon dopants and body dopants, collectively referred to as dopants **818** in the present application, are substantially co-implanted in the semiconductor layer to produce halo regions **852*a*** and **852*b*** having the first conductivity type.

[0085] Halo regions **852*a*** and **852*b*** are regions of semiconductor layer **816** comprising both carbon dopants and body dopants **818** having the first conductivity type. In one implementation, halo regions **852*a*** and **852*b*** may comprise carbon dopants and boron body dopants, and may have P-type conductivity. In the present implementation, the implantation energies and implant dosages of dopants **818** are configured such that halo regions **852*a*** and **852*b*** are situated at a depth below LDS **830*a*** and LDD **830*b*** and above BOX **814**, adjoining LDS **830*a*** and LDD **830*b***. In one implementation, during formation of halo regions **852*a*** and **852*b***, carbon dopants of dopants **818** may have an implant dosage of approximately $1.0*10^{15}/cm^3$, and may utilize an implantation energy of approximately fifteen kiloelectronvolts (25 keV), while body dopants of dopants **818** may have an implant dosage of approximately $4.0*10^{12}/cm^3$ to $1.0*10^{13}/cm^3$, and may utilize an implantation energy of approximately twelve to twenty kiloelectronvolts (12-20 keV). Halo regions **852*a*** and **852*b*** may have a higher concentration of body dopants having the first conductivity type compared to transistor body **844**. Halo regions **852*a*** and **852*b*** may have a lower concentration of body dopants having the first conductivity type compared to the concentration of dopants in LDS **830*a*** and LDD **830*b*** having the second conductivity type. For example, the concentration of P-type dopants in halo regions **852*a*** and **852*b*** may be greater than the concentration of P-type dopants in transistor body **844**, but less than the concentration of N-type dopants in LDS **830*a*** and LDD **830*b***.

[0086] Forming halo regions **852*a*** and **852*b*** may utilize an implant angled away from the direction normal to the top surface of semiconductor layer **816**. In various implementations, dopants **818** may be implanted at angle of approximately twenty to forty five degrees (20°-45°) from the direction normal to the top surface of semiconductor layer **816**, or greater or less. Thus, halo regions **852*a*** and **852*b*** have a different profile compared to LDS **830*a*** and LDD **830*b*** or a subsequently formed source or drain (not shown in FIG. **19**). For example, halo regions **852*a*** and **852*b*** may have a substantially elliptical or halo-shaped profile, rather than a substantially rectangular or rounded-rectangular profile.

[0087] In order to form halo regions **852*a*** and **852*b*** under both LDS **830*a*** and LDD **830*b***, in one implementation, the halo implant may maintain a fixed angle while SOI structure **808** may be rotated. In another implementation, the halo implant may comprise two implants performed at angles mirrored horizontally in FIG. **19**. In one implementation, forming each of halo regions **852*a*** and **852*b*** may utilize multiple halo implants at different angles. For example, forming each of halo regions **852*a*** and **852*b*** may comprise four successive halo implants at angles of twenty five degrees (25°), thirty degrees (30°), thirty five degrees (35°), and forty degrees (40°). As shown in FIG. **19**, halo regions **852*a*** and **852*b*** may be situated partially under gate **836**. In one implementation, blocking masks (not shown in FIG. **19**) may be situated over STIs **826** and/or portions of LDS **830*a*** and LDD **830*b***, in order to form halo regions **852*a*** and **852*b*** localized to areas under the sides of gate **836**.

[0088] FIG. **20** illustrates an SOI device processed in accordance with action **710** in the flowchart of FIG. **17** according to one implementation of the present application. As shown in FIG. **20**, a transistor is formed in semiconductor layer **816** in order to complete SOI device **810**, including source **828**a and drain **828**b having the second conductivity type and adjoining halo regions **852**a and **852**b.

[0089] In the implementation of FIG. **20**, halo regions **852**a and **852**b are illustrated as extending vertically from the top surface of semiconductor layer **816** to a depth between LDS **830**a or LDD **830**b and source **828**a or drain **828**b, and extending laterally from edges of source **828**a and drain **828**b to a width under gate **836**. In various implementations, halo regions **852**a and **852**b may have any other shapes or dimensions as they adjoin source **828**a and drain **828**b.For example, in one implementation, uppermost portions of halo regions **852**a and **852**b may be between the top surface of semiconductor layer **816** and bottom portions of LDS **830**a and LDD **830**b. In another implementation, lowermost portions of halo regions **852**a and **852**b may be below source **828**a and drain **828**b. In various implementations, source **828**a and drain **828**b may not extend to BOX **814**, or halo regions **852**a and **852**b may extend to BOX **814**. In yet another implementation, LDS **830**a and LDD **830**b may be optional.

[0090] Notably, because carbon dopants are substantially co-implanted with body dopants, halo regions **852**a and **852**b include both carbon dopants and body dopants. Body dopants in halo regions **852**a and **852**b inhibit punch-through between source and drain in short-channel devices. Meanwhile carbon dopants in halo regions **852**a and **852**b inhibits body dopants, such as boron, of halo regions **852**a and **852**b from diffusing upwards into channel region **832** and/or LDS **830**a and LDD **830**b during fabrication of SOI device **810** and during subsequent thermal processing, thereby reducing OFF state leakage current and other diffusion side effects. Except for differences noted above, SOI device **810** in FIG. **20** may generally correspond to SOI device **210** in FIG. **7**, and may have any implementations and advantages described above.

[0091] FIG. **21** illustrates a portion of a circuit including a radio frequency (RF) switch employing stacked SOI transistors according to one implementation of the present application. FIG. **21** represents an exemplary use case for SOI transistors, such as SOI device **210** in FIG. **7**, SOI device **606** in FIG. **16**, or SOI device **810** in FIG. **20**. The circuit in FIG. **21** includes RF switch **960** between first node **962** and second node **964**. First node **962** and second node **964** generally represent an input and an output respectively of RF switch **960**. In various implementations, first node **962** or second node **964** may correspond to an antenna, a filter, a low-noise amplifier (LNA), a power amplifier (PA), a ground, or another device.

[0092] RF switch **960** serves to pass or to block RF signals between first node **962** and second node **964**, depending on whether RF switch **960** is in an ON state or an OFF state. RF switch **960** includes a stack of SOI transistors **910**, which correspond to SOI transistors according to implementations of the present applications, such as SOI device **210** in FIG. **7**, SOI device **606** in FIG. **16**, or SOI device **810** in FIG. **20**. The drain **928**a of the first SOI transistor **910** is coupled to first node **962**, while subsequent drains **928**a are coupled to sources **928**b of previous SOI transistors **910**. The source **928**b of the last SOI transistor **910** is coupled to second node **964**. Gates **936** of SOI transistors **910** can be coupled to a controller or a pulse generator (not shown in FIG. **21**) for switching SOI transistors **910** between ON and OFF states. By stacking SOI transistors **910** as shown in FIG. **21**, the overall OFF state power and voltage handling capability for RF switch **960** can be increased. In various implementations, RF switch **960** may have more or fewer stacked SOI transistors **910** than shown in FIG. **21**. For example, RF switch **960** may have between six and thirty stacked SOI transistors **910**. In the present implementation, SOI transistors **910** are N-type field effect transistors (NFETs).

[0093] FIG. **22** illustrates an exemplary graph of body dopant concentration versus depth according to one implementation of the present application. In the graph, the y-axis represents the concentration of body dopants in a semiconductor layer after completion of the corresponding SOI device and any subsequent thermal processing. The x-axis represents the depth in the semiconductor layer, where depth D_1 corresponds to a top surface of the semiconductor layer, and depth D_4 corresponds to a bottom surface of the semiconductor layer. In the graph, dotted trace **966** represents the concentration of body dopants in the semiconductor layer of a conventional SOI device, while solid trace **968** represents the concentration of body dopants in the semiconductor layer of an SOI device according to the present application, such as an SOI device manufactured according to one of the flowcharts in FIGS. **1**, **8**, **13**, and **17**. Solid trace **968** may represent an exemplary vertical sample through a transistor body for an SOI device manufactured according to one of the flowcharts in FIG. **1**, **8**, or **13**, or an exemplary vertical sample through a halo region for an SOI device manufactured according the flowchart in FIG. **17**.

[0094] As shown by dotted trace **966** in FIG. **22**, body dopants have an initial body dopant concentration N_2 at depth D_1, which then rises to a peak body dopant concentration N_3 at depth D_2, which then falls to low body dopant concentration at approximately depth D_4. As shown by solid trace **968**, body dopants have an initial body dopant concentration N_1 at depth D_1, which then rises to a peak body dopant concentration N_4 at depth D_3, which then falls to low body dopant concentration at approximately depth D_4.

[0095] As shown by dotted trace **966** and solid trace **968** in FIG. **22**, near the top of the semiconductor layer (i.e., between depth D_1 and depth D_2), the conventional SOI device has a higher body dopant concentration than the SOI devices according to the present application, whereas near a bottom portion of the semiconductor layer (i.e., between depth D_2 and depth D_4), the SOI devices according to the present application have a higher body dopant concentration than the conventional SOI device. In one implementation, the area under each of dotted trace **966** and solid trace **968**, or the total body dopants is approximately equal. The difference in shape between dotted trace **966** and solid trace **968** represents body dopants having diffused upwards in the conventional SOI device. The SOI devices according to the present application effectively inhibited such diffusion by introducing carbon dopants to body regions, as described above. Accordingly, the SOI devices according to the present application effectively inhibit detrimental side effects of such diffusion, and can also maintain a retrograde doping profile in the semiconductor layer with respect to body dopants having the first conductivity type. In various implementations, traces **966** and **968** may exhibit different patterns or slopes than shown in FIG. **22**.

[0096] SOI structures according to the present invention, such as SOI device **210** in FIG. **7**, SOI structure **408** in FIG. **11**, SOI device **606** in FIG. **6**, and SOI device **810** in FIG. **20**, result in numerous advantages, some of which are stated below. First, because carbon dopants inhibit body dopants, such as boron, from diffusing upwards during fabrication of SOI devices and during subsequent thermal processing, SOI structures according to the present invention maintain a higher concentration of body dopants lower in the semiconductor layer. This advantageously inhibits short-channel effects that would otherwise occur, such as punch-through between the source and the drain, especially where the higher concentration of body dopants extends to the critical interface of the semiconductor layer with the BOX.

[0097] Second, SOI devices according to the present invention experience lesser gate induced drain leakage (GIDL) current in an OFF state compared to typical SOI devices. Accordingly, an RF switch such as RF switch **960** in FIG. **21** employing SOI transistor **910** will experience overall higher power handling in an OFF state. Where RF switch **960** requires high OFF state power handling capability and employs numerous stacked SOI transistors, the advantages of the higher OFF state power handling capability of SOI transistor **910** are compounded.

[0098] Third, SOI structures according to the present invention enable a variety of dopant profiles, including a retrograde body doping profile, in the semiconductor layer. As a result, as described above, SOI devices benefit from a significantly lower R_{ON} through the channel region, the LDS, the LDDs, the source silicide, and the drain silicide. Moreover, SOI devices benefit from a significantly lower product of ON-state resistance and OFF-state capacitance ($R_{ON} \times C_{OFF}$), without trading off breakdown voltage handling capability.

[0099] Fourth, SOI structures according to the present invention significantly reduce the short-channel effect of threshold voltage roll-off. Accordingly, device dimensions such as channel length and gate length can be scaled down with fewer performance trade-offs.

[0100] Fifth, SOI structures according to the present invention can be formed from typical pre-fabricated SOI substrates, such as SOI structure **202** shown in FIG. **2**. Accordingly, SOI structures according to the present invention are suitable for large scale production without premium costs associated with specialty substrates.

[0101] From the above description it is manifest that various techniques can be used for implementing the concepts described in the present application without departing from the scope of those concepts. Moreover, while the concepts have been described with specific reference to certain implementations, a person of ordinary skill in the art would recognize that changes can be made in form and detail without departing from the scope of those concepts. As such, the described implementations are to be considered in all respects as illustrative and not restrictive. It should also be understood that the present application is not limited to the particular implementations described above, but many rearrangements, modifications, and substitutions are possible without departing from the scope of the present disclosure.

1-11. (canceled)

12. A method of fabricating a semiconductor-on-insulator (SOI) device, said method comprising:
- providing a semiconductor layer over a buried oxide, said buried oxide over a handle wafer;
- forming a transistor body having a first conductivity type in said semiconductor layer;
- forming a gate over said transistor body;
- substantially co-implanting carbon dopants and body dopants in said semiconductor layer to produce a halo region having said first conductivity type;
- forming a source and a drain having a second conductivity type, at least one of said source and said drain adjoining said halo region.

13. The method of claim **12**, further comprising forming at least one of a lightly-doped source and a lightly-doped drain in said transistor body, wherein said halo region adjoins said at least one of said lightly-doped source and said lightly-doped drain.

14-21. (canceled)

22. The method of claim **12**, wherein a peak concentration of said carbon dopants is at a first depth, and wherein a peak concentration of said body dopants is at a second depth below said first depth.

23. The method of claim **22**, wherein said source and said drain are situated below said first depth.

24. The method of claim **22**, wherein said lightly-doped source and said lightly-doped drain are situated above said first depth.

25. The method of claim **12**, wherein said body dopants have P-type conductivity.

26. The method of claim **25**, wherein said boy dopants having P-type conductivity comprise boron.

27. The method of claim **25**, wherein said body dopants having P-type conductivity are formed using difluoride (BF_2) with an implantation energy of approximately fifteen kiloelectronvolts (15 keV).

28. The method of claim **12**, wherein said source and a drain comprise N-type dopants.

29. The method of claim **13**, wherein said lightly-doped source and said lightly-doped drain comprise N-type dopants.

30. A method of fabricating a semiconductor-on-insulator (SOI) device, said method comprising:
- providing a semiconductor layer over a buried oxide, said buried oxide over a handle wafer;
- forming a transistor body having a first conductivity type in said semiconductor layer;
- forming a gate over said transistor body;
- forming a lightly-doped source and a lightly-doped drain having a second conductivity type;
- substantially co-implanting carbon dopants and body dopants in said semiconductor layer, wherein a peak concentration of said carbon dopants is at a first depth, and a peak concentration of said body dopants is at a second depth below said first depth;
- forming a source and a drain having said second conductivity type.

31. The method of claim **30**, further comprising forming a halo region having said first conductivity type in said semiconductor layer.

32. The method of claim **31**, wherein said halo region includes said carbon dopants and said body dopants, and wherein at least one of said source and said drain adjoins said halo region.

33. The method of claim **31**, wherein said halo region adjoins at least one of said lightly-doped source and said lightly-doped drain.

34. The method of claim **30**, wherein said source and said drain are situated below said first depth.

35. The method of claim **30**, wherein said lightly-doped source and said lightly-doped drain are situated above said first depth.

36. The method of claim **30**, wherein said body dopants have P-type conductivity.

37. The method of claim **36**, wherein said body dopants having P-type conductivity comprise boron.

38. The method of claim **30**, wherein said source and a drain comprise N-type dopants.

39. The method of claim **30**, wherein said lightly-doped source and said lightly-doped drain comprise N-type dopants.

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